LICENTIATE THESIS

Modeling, Control and Path Planning for an Articulated Vehicle

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Modeling, Control and Path Planning for an Articulated Vehicle

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ABSTRACT

This licentiate thesis is focusing on the modeling, control and path planning problems for an articulated vehicle. Based on a novel error dynamics modeling approach, the nonlinear kinematic model of the vehicle has been transformed in a linear switching model representation, which is also able to take under consideration the effect of slip angles, a factor that significantly deteriorates the overall system approach.

According to the proposed switching modeling, a switching model predictive control scheme has been established. Each of the controllers has been fine tuned for dealing with a specific model representation, translational speed, magnitude of slip angles and physical / mechanical constraints. In the proposed control scheme the varying slip angle and speed have been considered as the switching signal for selecting the active controller.

The final contribution is related to the path planning problem where an on line path planning algorithm, being able to consider the full kinematic model of the vehicle, have been proposed. The presented algorithm is a variation of the well known Bug-like path planning scheme. The efficiencies of the presented approaches have been evaluated by multiple simulation results.

CONTENTS

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First and foremost thanks to Allah, I praise God for the blessings endowed upon me and provide me perseverance and patience to complete the work entrusted to me. My deep sense of gratitude is expressed here to my supervisors: George Nikolakopoulos and Thomas Gustafsson for their valuable supervision, guidance and enlightening discussions throughout the period of my work and preparation for this licentiate thesis. I am indebted to my wife and my kids, without their help, encouragement and patience I would never have gotten this thesis completed. I am deeply thankful to my beloved mother (especially she is away around 5000Km) who brought me up with endless love, perfect care and great expectations. Finally, special thanks goes to all the other people who have supported me during the courses and the research, (especially Khalid T. Atta) and all my friends who provided help, suggestions and good accompany in the long days that I have spent in my office.

Thaker Nayl, 2013

Part I

CHAPTER 1

Introduction

1.1 Background

Recently, there has been significant advances in designing automated vehicles, with a general aim to increase the technological readiness in autonomous operations. The theory and application of autonomous vehicle are currently among the most intensively studied and promising areas in control engineering. The load haul dump (LHD) vehicle in Figure 1.1, is a typical example of an articulated vehicle.

The overall main parts of research presented in this licentiate concern: a) modeling, b) control and c) path planning for an articulated vehicle. More analytically the main parts of the thesis, with the corresponding objectives are as follows:

- **Part 1: Modeling and Control:** 1) Two non–linear kinematic models for an articulated vehicle have been considered, 2) comparison of the overall complexity of the proposed models has been made, 3) transformation of these models into an error dynamics model, which in the sequence are being linearized around multiple nominal operation points, 4) based on the derived multiple error dynamic models, the varying slip angle is being considered as the switching rule for the switching control scheme, and 5) switching linear model predictive control is being proposed to take under considerations: a) multiple constraints, b) model uncertainties, and c) switching in the system modeling.

- **Part 2: Collision Avoidance Algorithm:** A novel online dynamic smooth path planning and collision avoidance scheme based on a bug-like modified path planning algorithm for an articulated vehicle has been done in the research. This algorithm is based on the limited range sensing of the surrounding environment. It has the following properties: a) on line collision avoidance path planning scheme based on a bug-like modified path planning algorithm, b) the path planning based on the real dynamic model under limited and sensory reconstructed surrounding static environment, and c) low computational complexity.

- **Part 3: Combine Path Planning with Model Predictive Control:** Combined situations of control feedback and on line path planning algorithms moving in a partially known and sensory based reconstructed environment have been considered. A model predictive controller is being utilized for creating the proper control signal at the same time with path planning. Path planning algorithms with the capability to take under consideration the real dynamics of the articulated vehicle, based on a priori knowledge of the current and the goal points and a partial sensory based awareness of the surrounding environment have been proposed.

Figure 1.1: Typical articulated vehicle

1.2 Research Goal

The proposed research is trying to meet the following objectives: build an articulated vehicle model under taking the effect of the slip angle into account, achieve the stable motion by using single and multi-controllers, follow a path, arrive to the final goal, avoid collision, update path on line, achieve a short distance and small control effort. Figure 1.2, shows this sequence of these objectives. The advantages of the articulated vehicle can be explained in two terms: a) degrees of freedom in the selection of the directions needed to span the whole tangent space and b) reduce the gap between the trajectories followed by the rear part of the vehicle when applied a steering.

Theoretically, autonomous control of the articulated vehicle with situations of control feedback and path planning algorithms of an articulated vehicle has been performed in the research literature. Model Predictive Control (MPC) is successful and aware of actuator limitations, allows operation close to constraints and flexible specification of objectives. This research has dealt with the static environment case and local path planning problem. The proposed algorithm allows a mobile vehicle to navigate through static obstacles, and finding the path in order to reach the goal without collision. Calculated the minimum total distance of the path from the origin to the destination with and without obstacles in the environment. Also the vehicle could be passing to avoid dynamic obstacles that are moving in time.

The novelty of this research can be summarizing in the following objectives:

a) derivation of an error dynamics modeling framework for the case of an articulated vehicle operating under varying slip angles, b) adopting a switching model control scheme that is able to neglect the effect of slip angles, c) consideration the effects of real life constraints on the control input and mechanical restrictions, d) proposed of a new bug-like path planning algorithm based on the kinematic model of an articulated vehicle, and e) providing online collision avoidance and optimal or sub-optimal solutions.

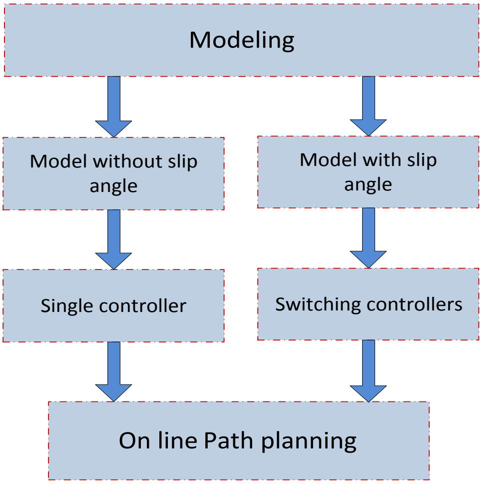

Figure 1.2: Research sequence for control and path planning a robot

The evaluation of the research is presented in terms of performance of restricted mobility, different limitation on control steering angle, range sensor, safe distance dimensions and total reaching distance. The research contains an examined the relationship between the vehicle geometric and the desired path, and using it to derive a mathematical, linear state space model of the vehicle to follow a path and reach the final goal point.

1.2.1 Modeling

Many researches discussed the steering kinematics for a center articulated mobile vehicle using different models that have been compared and confirmed by empirical data [1, 2]. A simple dynamic model for the planar motion of a LHD truck, using Lagrange equations with incorporating kinematic constraints to approximate the forces can found in [3]. Moreover, authors in [4], provided a summary of the derivation of a dynamic model of a LHD vehicle using Lagrangian dynamics alongside with a simplified tire model in a try to include the effect of basic tire dynamics. The vehicle which is considered in this research belongs to the category of articulated vehicles that the kinematic joints can be described as algebraic constraint equations [5].

Researches has been conducted trying to model, simulate and control vehicles with focus on explaining the nature of their dynamic response [6–8]. They also derived an expression to show the relationship between the vehicle speed and stability, which is used to optimize the dynamic response of the vehicle by constituting a foundation for an adaptive tuning approach. In this research, the existence of the slip angles has been presented. Slip angles have a significant effect on the vehicle's path tracking capability and can significantly deteriorate the performance of the overall control scheme.

1.2.2 Controlling

From a control point of view, many traditional techniques for non–holonomic vehicles, based on error dynamics models without the presence of slip angles have been proposed. More analytically, in [9, 10] linear control feedback has been applied. A controller that consist of two parts to solve the path tracking problem for LHD mining vehicle based on a geometric approach has been presented in [11]. In [12], kinematic tracking controller for mobile vehicles with kinematic uncertainty which combines input-output linearization method and Neural Network control with adaptive and robust control techniques.

In [13], specific criteria have been presented for articulated vehicle truck to track a path presented by a line crossing the tunnel center of an underground mine. A mobile vehicle may be strongly controllable in its configuration space, as established by various authors focusing on path planning issues [14–16] and path planning using a kinematic vehicle model presented in [17, 18].

A general common vehicle stability system can be controlled by active steering [19]. Feedback control of non-holonomic mobile vehicles is currently arising the interest of several research groups, [20–23]. A variety of problem types defined in the literature can be founded in [24]. This research suggested a switching model predictive control scheme for controlling an articulated vehicle under varying slip angles. A corresponding switching mode predictive control scheme based on the varying slip angles and the derived multiple error dynamic models.

1.2.3 Vehicle Motion Planning

The aim of path planning algorithm for autonomous mobile robots is to search a collision free path between a starting and a final goal point. These algorithms provide the robot with the possibility to move from the initial position to the goal position.

In the modified path planning algorithm the vehicle is always searching for the shortest distance to the goal point. Optimality may be defined as minimizing the time between initial and goal points, or by minimizing some other attributes of the path for example, distance and energy consumed. Then, the optimality is the shortest distance with collision free path under some restrictions. It is possible to further categorize problems based on the assumed vehicle shape, environment type and behavior. A brief overview of motion planning can be static or dynamic, depending on the mode in which the obstacle information is available.

Several approaches for path planning exist for mobile vehicles, whose suitability depends on complexity of the environment, sensors and vehicle capabilities. Depending on the environment where the vehicle is located in, the path planning can be categorised into the following two types: a) static environment which only contains the static obstacles, and b) dynamic environment which has static and dynamic obstacles. Each of these two types could be further divided in two sub groups, as in Figure 1.3.

In a static problem, all the information about obstacles is known a priori, and the motion of the vehicle is designed from the given information. A problem is considered static if there is perfect knowledge of the environment, and dynamic if knowledge of the environment is imperfect or changes as the task unfolds. In dynamic planning, only partial information is available about the obstacles.

Path planning typically refers to the design of only geometric (kinematic) specifications of the positions and orientations of vehicles. Consistently keeping a safe distance from obstacles, and producing smooth paths that exhibit desirable properties are typical requirements. This is used to update the internal representation of the vehicle's environment, and the vehicle replans a path from the updated representation. The process of updating the representation and local planning paths is continued until the vehicle accomplishes its goal.

Path planning for any robots or vehicle has been categorized into two types: Off line algorithms require a priori a map of the environment. The path is pre-computed and then given to the vehicle to execute. The vehicle uses this path information to navigate itself efficiently through the environment with the help of a path planning algorithm.

On line algorithm is used to avoid obstacles by reacting to data collected from onboard sensors. It can be used when a map of the environment is not known or, if an unexpected obstacle was encountered during the execution of a pre-computed path. Since on line path planners run in real time and during the vehicle movement towards the goal point. The quality of the on line planning depends on the look-ahead distance of the sensor.

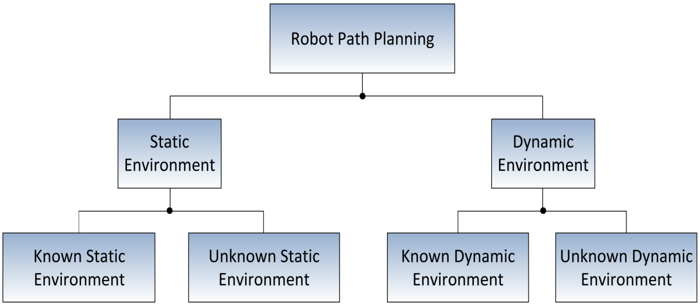

Figure 1.3: Schematic representation of path planning algorithms

In this licentiate thesis, the problem of path planning for unknown dynamic environment has been addressed, where it is being assumed that the robot knows its starting and final goal points.

1.3 Vehicle and Environment Limitations

There are various external limitation factors that degrade the system performance and introduce errors in the model that propagate with time. These errors that could also be considered as external disturbances in the system model, are being caused by many factors, such as: a) the generic interaction between the vehicle and its environment [25], b) the noise and bias in positioning and driving sensors [26], different sensor devices can basically act on mobile vehicles in functionality, c) existence or variation of slipping angles among front and rear parts depending on the type of the driving ground, as well as the fatigue and type of the tyres [27], and d) dynamic effects resulting from acceleration and braking [13].

Slip angle is one of the most important factors that degrades the overall performance of the articulated vehicle. When the dynamics of the vehicle are being incorporated the motion and the overall behavior of the vehicle significantly deviates from the case where the vehicle is being considered as having the dynamics of an unconstraint point and this is one of the major contributions of this thesis. Moreover slip angle is an important state variable in vehicle dynamics. It's one of the most crucial contributor components in automation of the vehicles dynamics and for vehicle motion control [28]. The slip angle defined as an angular difference between the direction of the tyre contact patch with the road and the direction of the wheel.

There are two types of slip angles lateral and longitudinal, and of this research only considers the lateral (side) slip angles. In the lateral sort, there are two slip variables β and α, being defined as the front's and rear's slip angles respectively.

In most of the presented path planning algorithms, the vehicle is being modeled as a point within the world space, without any constraint in the movements, while the actual kinematics of the vehicle are been neglected. The actual kinematic is important especially in the case of non–holonomic vehicles. The main disadvantage of a point vehicle is the fact that it is considered to be a single point, without any constraints in the movement, denoted as $\dot{x} = u$, where u is the control action.

A kinematic constraint is a non-holonomic constraint if it can be expressed in the form [29]:

$$f(\mathbf{X}, \theta) = 0 \tag{1.1}$$

In problems with differential constraints, time and states have to satisfy the equations of motion of the vehicle. Typically the states are constrained by the limitation on the control steering angle acceleration and velocity. The velocity is constrained by the following relation: $0 \leq v \leq v_{max}$.

1.3.1 Environments

The general idea is that a path is planned based on the environment known right now. When the vehicle gets new information it considers choosing another path. Based on an a priori knowledge of the current and the goal points, and a partial sensory based awareness of the surrounding environment.

If the knowledge of the environment is known, the global path can be planned before the vehicle start to moves. This global path can assist the vehicle to traverse within the real environment because the feasible optimal path has been constructed within the environment. The table below, shows the difference between the two mobile vehicle navigation global and local approaches, [30].

Mobile vehicle path planning approaches

Local navigation	Global navigation
-Information based on local sensors	-Map based
-Incomplete knowledge of environment	-Complete knowledge of environment
-On line avoiding obstacles while moving to goal	-Obtain off line a feasible path to goal
-Reactive system	-Deliberative system
-Fast response	-Slow response

Characteristic cases of the global navigation are the road–map algorithm, the cell decomposition, the Voronoi diagrams, the occupancy grinds and the new potential fields techniques, while in most of the cases, a final step of smoothing the produced paths curvatures, by the utilization of Bezier curves is being utilized. The disadvantage of these methods are complexity and the cost of the sensors and time consuming.

Local path planning is a partially known and on line reconstructed environment, the Bug-family algorithms are well known mobile vehicle navigation methods for local path planning based on a minimum set of sensors and with a decreased complexity for on line implementation. Two of the most commonly utilized path planning algorithms in this category are the Bug1 and Bug2. This algorithm is the simplest and most applicable in reality. Moreover, it is assumed that the vehicle is constantly aware of the final goal coordinates. During the convergence to this goal and based on the limited sensing range of the surrounding environment, the vehicle is able to detect and avoid obstacles, while continue converging to the optimum goal. This approach is providing an online and suboptimal solution, when compared with the global path planning techniques, and it can be directly applied to the case of articulated vehicles.

1.4 Thesis Outline

The organisation of the thesis is as follows: In the first part, Chapter 1, contains an introduction to the problem and additional background information about the autonomous vehicle and its limitations. Chapter 2 introduces basic concepts of modeling, including the classical kinematic model and then the kinematic model with slip angels. Background theory also is also provided in this chapter with respect to the suggested control scheme and the path planning algorithm approach.

The second part, contains the four research papers related to this thesis. First, full non-linear kinematic models with the ability to take the slipping angles under consideration, with multiple parameters sets for slip angles and different steering angles. The next two papers, present a switching model predictive control scheme for an articulated vehicle under varying slip angles and different velocities. The existence of the slip angles has a significant effect on the vehicle's path tracking capability and can significantly deteriorate the performance of the overall control scheme. In the last paper, a new online and sub–optimal bug-like path planning algorithm for the case of an articulated vehicle, utilized to fined collision free path and generate smooth control points using model predictive control in on line mode is being presented.

Chapter 2

Modeling, Control and Path Planning

Several kinematic models have been recently proposed in the scientific literature, for articulated vehicle with two or multiple body dynamics. Most of these methods contain simple models that are not taking the effect of the slipping angles under consideration, as in this case the problem is being complicated and it is more difficult to proceed to the next stage of the control development.

2.1 Modeling of an Articulated Vehicle

In general an articulated vehicle consist of two parts, a front and a rear with a width of w and lengths l_1 and l_2 correspondingly, linked with a rigid free joint. In modeling a mechanical system, the links and bodies may be interconnected by one or more kinematic joint, while each body has a single axle and the wheels are all non–steerable. The centers of gravity for these parts are being denoted as (x_1, y_1) and (x_2, y_2) respectively. The steering action is being performed on the joint by changing the corresponding articulated angle γ between the front and the rear of the vehicle, as it is being indicated in Figure 2.1. The front and rear parts in an articulated vehicle model are connected together by revolute joint, which is a lower pair kinematic joint that does not need any information on the shape of the connected bodies and can be expressed by two constraints. In case that the non-holonomic constraints acting on the front and rear axles, are taken into consideration, mainly resulting from the assumption of rolling without slipping movement for the wheels of the vehicle. These can be expressed as it follows:

$$\dot{x}_1 \sin\theta_1 - \dot{y}_1 \cos\theta_1 = 0 \tag{2.1}$$

$$\dot{x}_2 \sin\theta_2 - \dot{y}_2 \cos\theta_2 = 0 \tag{2.2}$$

For deriving the vehicle's kinematic equations, it is assumed that a) the steering angle γ remains constant under small displacement, b) the dynamical effects due to low speed (like tire characteristic, friction, load and breaking force) are being neglected, c) each axle is being composed of two wheels that can be replaced by a unique wheel, and d) the vehicle moves on a plane without slipping effects. The vehicle's velocity is bounded within the maximum allowed velocity, which prevents the vehicle from slipping. From the geometrical characteristics of the vehicle it can be easily derived that the

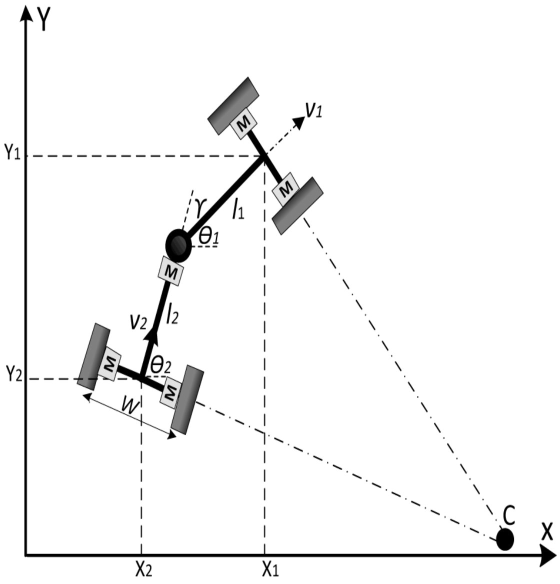

Figure 2.1: Articulated vehicle's geometry and graphical representation of dynamics coordinates

equations describing the translation of the articulated front part are:

$$\dot{x}_1 = v_1 \cos \theta_1 \tag{2.3}$$

$$\dot{y}_1 = v_1 \sin \theta_1 \tag{2.4}$$

The velocities v_1 and v_2 are considered to have the same changing with respect to the velocity of the rigid free joint of the vehicle and thus the relative velocity vector equations can be defined as:

$$v_1 = v_2 \cos \gamma + \dot{\theta}_2 l_2 \sin \gamma \tag{2.5}$$

$$v_2 \sin \gamma = \dot{\theta}_1 l_1 + \dot{\theta}_2 l_2 \cos \gamma \tag{2.6}$$

where $\dot{\theta}_1$ and $\dot{\theta}_2$ are the angular velocities of the front and rear parts respectively. Combining equations (2.5) and (2.6) yields:

$$\dot{\theta}_1 = \frac{v_1 \sin \gamma + l_2 \dot{\gamma}}{l_1 \cos \gamma + l_2} \tag{2.7}$$

For the case that there is a steering limitation for driving the rear part according to the coordinates of the point (x_2, y_2), the relationship between front and rear coordinates is provided by:

$$x_2 = x_1 - l_1 \cos \theta_1 - l_2 \cos \theta_2 \tag{2.8}$$

$$y_2 = y_1 - l_1 \sin \theta_1 - l_2 \sin \theta_2 \tag{2.9}$$

In the presented approach, the manipulated variable is the rate of the articulated angle, while the vehicle's speed is considered to be constant, and the vehicle's kinematic model can be presented in a general form as:

$$\dot{\mathbf{x}} = f(\mathbf{x},\mathbf{u}) \tag{2.10}$$

The realistic dynamic motion behavior of the articulated vehicle with initial parameters $[x\ y\ \theta\ \gamma]$, is depicted in Figure 2.1. The state parameters of the articulated vehicle are; $\mathbf{X} = [X\ Y\ \theta\ \gamma]^T$ and the manipulated variables are $\mathbf{u} = [v\ \dot{\gamma}]^T$, while the kinematic model of the articulated vehicle, in a state space can be written as it follows:

$$\begin{bmatrix}\dot{X}\\ \dot{Y}\\ \dot{\theta}\\ \dot{\gamma}\end{bmatrix} = \begin{bmatrix} cos\theta & 0\\ sin\theta & 0\\ \frac{sin\gamma}{l_1 cos\gamma + l_2} & \frac{l_2}{l_1 cos\gamma + l_2}\\ 0 & 1\end{bmatrix}\begin{bmatrix} v\\ \dot{\theta}_\gamma\end{bmatrix} \tag{2.11}$$

where $\dot{\theta}_\gamma$ is the rate of change for the articulated angle. The angular velocity of the rear part can be write as:

$$\dot{\theta}_2 = \frac{v_1\ \sin\gamma - l_1\ \dot{\gamma}\cos\gamma}{l_1\ \cos\gamma + l_2} \tag{2.12}$$

The corresponding angular velocity for the front part is θ_1 and for rear parts is θ_2, and have different values when the vehicle has different lengths.

2.1.1 Kinematic Model Under Slip Angles

In the case of slip angles, the kinematic model of the articulated vehicle can also be formulated from general geometry that includes ideal and slip behavior for the steering angles. The definition of this model has been initially based on the derivation in [31], which included the perturbed factors in the vehicle position as two slip variables β and α. Under the influence of slip angles, the vehicle's configuration is depicted in Figure 2.2, where $(r_1,\ r_2)$ are the radius from instantaneous centers of velocity for the front and the rear parts respectively. For the initial center curvature of the trajectory, is assumed that the vehicle is moving forward without slip conditions.

In the following derivation, the unit subscript 's' denotes variables in the slip angle case as they have been defined previously in the non-slip examined case. The kinematic equations, for the motion under the effect of slip angles, for the front part can be formulated as:

$$\dot{x}_{1s} = v_{1s}\cos\left(\theta_1 + \beta\right) \tag{2.13}$$
$$\dot{y}_{1s} = v_{1s}\sin\left(\theta_1 + \beta\right) \tag{2.14}$$

In this case, the vehicle's motion depends not only on vehicle's velocities, the articulated angle and the vehicle's lengths, but also on the slip angles, while the resulting vehicle's heading is provided by a combination of the slip angles and the orientation angles. The rate of the orientation $\dot{\theta}_{2s}$ can be defined as function of the steering angle γ and both slip angles. Based on the assumption that the vehicle develops a steady-state motion turn, this rate can be provided by the utilization of a virtual center of rotation, depending on the velocity. In the examined case, the front's and rear's speeds can be computed as:

$$v_{1s}\cos\alpha = v_{2s}\cos(\gamma - \alpha) + \dot{\theta}_2 l_2\ \sin(\gamma + \beta - \alpha) \tag{2.15}$$
$$v_{2s}\ \sin(\gamma + \beta - \alpha) = \dot{\theta}_1 l_1\ \cos\alpha + \dot{\theta}_2 l_2\ \cos(\gamma - \alpha) \tag{2.16}$$

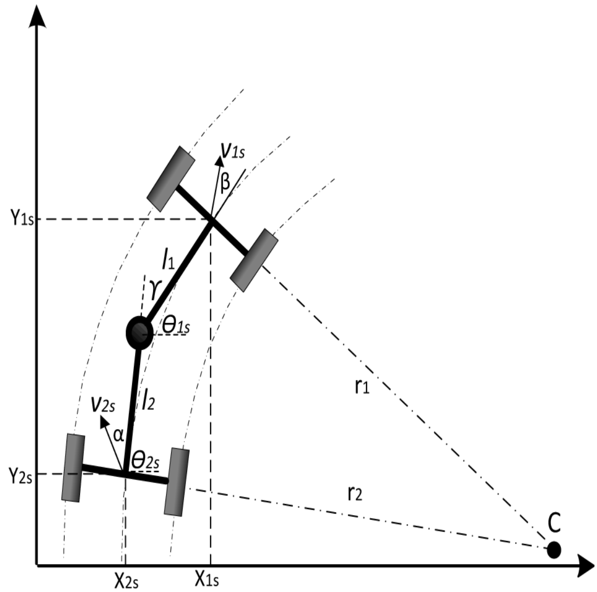

Figure 2.2: Articulated vehicle modeling configuration under the influence of slip angles

while, by solving the above equation, the angular velocity of the front can be defined as:

$$\dot{\theta}_{1s} \quad = \quad \frac{v_{1s}\ \sin(\gamma+\beta-\alpha)+l_2\ \dot{\gamma}\ \cos\alpha}{l_1\ \cos(\gamma-\alpha)+l_2\ \cos\alpha} \tag{2.17}$$

It should be noted that equation (2.17) is more accurate than equation (2.7), if the slip angles are known a priori, a task that is quite difficult as the slip angles are dependent of the vehicle's velocity, mass, tire-terrain interaction and articulation angle, in a highly non-linear way. In a state space formulation, this relation can be expressed as it follows:

$$\begin{bmatrix} \dot{X}_s \\ \dot{Y}_s \\ \dot{\theta}_s \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} \cos(\theta+\beta) & 0 \\ \sin(\theta+\beta) & 0 \\ \frac{\sin(\gamma+\beta-\alpha)}{l_1\cos(\gamma-\alpha)+l_2\cos\alpha} & \frac{l_2\cos\alpha}{l_1\cos(\gamma-\alpha)+l_2\cos\alpha} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \dot{\theta}_\gamma \end{bmatrix} \tag{2.18}$$

2.2 Error Dynamics Modeling

Before proceeding with the design of the control scheme, an appropriate the kinematic model should be initially derived. In the sequel, the error dynamics modeling procedure will be presented, while it should be highlighted that this is the first time in the relevant scientific literature that such a modeling framework, with the ability to take under consideration the effect of slip angles, has being presented.

Based on the assumption that the vehicle develops a steady–state motion turning, or $\dot{\gamma}=0$, the rates of the orientation change are being provided by: $r_1=\frac{v_1}{\dot{\theta}_f}$ and $r_2=\frac{v_2}{\dot{\theta}_r}$. In Figure 2.3 it is depicted an overview of how the curvature, heading and displacement errors are defined, between the actual path of the articulated vehicle and the desired one. In this Figure, the distance from the vehicle to the reference path displacement, as the angle between the vehicle and the reference are also being displayed.

In the following derivation, three errors are defined as: a) e_c is the curvature error, b) e_h is the heading error, and c) e_d is the displacement error. The curvature can be presented using the equation $1/r - 1/R$, heading error is the angle between the lateral axis of the front part and the line that connects between midpoint of front axle and the circular path center. Finally the displacement error e_d is the difference between the coordinates of the tractor and the coordinates of the desired circular path. By

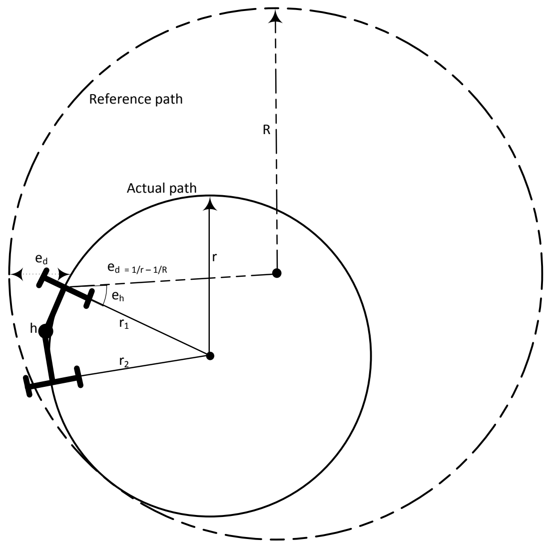

Figure 2.3: A graphical representation of the error dynamics transformation

defining an error state vector as $\dot{\mathbf{x}}_e = [\dot{e}_c \;\; \dot{e}_d \;\; \dot{e}_h]^T$, the error dynamics description of the vehicle can be presented as it follows [32]:

$$\begin{aligned} \dot{e}_c &= k_1 \, \dot{\gamma} \\ \dot{e}_h &= v \, e_c + k_2 \, \dot{\gamma} \\ \dot{e}_d &= v \, e_h \end{aligned} \tag{2.19}$$

where, $k_1 = \frac{1}{l_1+l_2}$ and $k_2 = \frac{l_2}{l_1+l_2}$ are constants. In this formulation it has been assumed that the three state variables are measured and that direct measurements for both the velocity and the position of the vehicle can be performed based on the on-board sensory system and thus no estimation action should be applied to the system.

2.3 State Space Model Under Slip Angles Effects

In case of slip angles, the kinematic model of the articulated vehicle can be also formulated from general geometry that includes slip behavior for the steering angles. The rate of the orientation can be defined as function of the steering angle γ and both slip angles. Based on the assumption that the vehicle develops a steady state motion turn, this rate can be provided by the using a virtual center of rotation, depending on the velocity.

With respect to the local coordinate axis origin and with the assumption that the articulated angle is changing at a constant rate with forward velocities and slip angles for the two parts of the vehicle, the velocities at the rigid free joint of the vehicle are equal to the front velocity v_1 rotated by γ degrees that for the front part and equal v_2 rotated by γ degrees for the rear part, while adding the effects of slip angles at each part, the lengths and the angular velocities. By obtaining the velocity of the point $p(x_1, y_1)$ in the direction perpendicular to v_1 we obtain the relation:

$$\begin{aligned} v_1\, cos\alpha &= v_2\, cos(\gamma-\alpha)+\dot{\theta}_2 l_2\, sin(\gamma+\beta-\alpha) && (2.20)\\ v_2\, sin(\gamma+\beta-\alpha) &= \dot{\theta}_1 l_1\, cos\alpha+\dot{\theta}_2 l_2\, cos(\gamma-\alpha) && (2.21)\end{aligned}$$

The solution to the above equations in order to calculate the angular velocity of the front part is:

$$\dot{\theta}_1 = \frac{v\, sin(\gamma+\beta-\alpha)+l_2\, \dot{\gamma}\, cos\alpha}{l_1\, cos(\gamma-\alpha)+l_2\, cos\alpha} \tag{2.22}$$

With the assumption that the slip angles, the velocity and the curvature of the trajectory are piecewise constant and by differentiating the equation: $e_c = \frac{v\, sin(\gamma+\beta-\alpha)+l_r\, \dot{\gamma}\, cos\alpha}{v\, (l_f\, cos(\gamma-\alpha)+l_r\, cos\alpha)}$ with respect to time, the rate of the curvature error is defined as:

$$\begin{aligned} \dot{e}_c = \quad & \dot{\gamma}\frac{l_f\, cos(\beta)+l_r\, cos(\alpha)\, cos(\gamma+\beta-\alpha)}{(l_f\, cos(\gamma-\alpha)+l_r\, cos(\alpha))^2}+ \\ & \ddot{\gamma}\,\frac{l_r\, (l_f\, cos(\gamma-\alpha)\, cos(\alpha)+l_r\, cos(\alpha)^2)}{v(l_f\, cos(\gamma-\alpha)+l_r\, cos(\alpha))^2}+\dot{\gamma}^2\, \frac{l_f\, l_r\, sin(\gamma-\alpha)\, cos(\alpha)}{v(l_f\, cos(\gamma-\alpha)+l_r\, cos(\alpha))^2}\end{aligned}$$

As a last step, the following state space error dynamics description for the articulated vehicle after linearization is being extracted:

$$\begin{bmatrix}\dot{e}_d\\ \dot{e}_h\\ \acute{e}_c\end{bmatrix} = \begin{bmatrix}0 & v & 0\\ 0 & 0 & v\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}e_d\\ e_h\\ \acute{e}_c\end{bmatrix} + \begin{bmatrix}\frac{l_f\beta}{l_r+(\beta-\alpha)(l_f+l_r)}\\ \frac{}{(l_f+l_r)}\\ \frac{1}{(l_f+l_r)}\end{bmatrix}\dot{\gamma} \tag{2.23}$$

In the case that the articulated vehicle is driven with varying speed, over terrains that are characterized by different slip angles, multiple piecewise operating sets for the speed and the slip angles, should be considered as it is shown in [*PaperC*].

2.4 Model Predictive Control

Model predictive control is used for creating the proper control signal, the rate of the articulated angle based on error dynamics kinematic model of the vehicle. MPC has been very successful in practice and its a highly effective control scheme that is able to take under consideration multiplicative system model descriptions, uncertainties, nonlinearities and physical and mechanical constraints in the system model parameters or in the control signals [33].

The MPC action, which is the rate of the articulation angle, is based on a finite horizon continuous time minimization of predicted tracking error with constraints on the control inputs and the state variables, a step after modeling and design controllers in this thesis is the proposed of a path planning algorithm based on the error dynamics kinematic model has the ability to consider the physical constraints of the vehicle while avoiding obstacles. The switching adaptive MPC, as it has been presented in [*PaperB*], can deal with the uncertainty imposed on the kinematics model of mobile vehicles by selecting the exact controller. This uncertainty comes from the effect of varying slip angles.

The overall block diagram of the proposed closed loop system is depicted in Figure 2.4. For efficiently controlling the articulated vehicle, the controller utilizes the current state of motion of the vehicle as well as the next target points of the reference trajectory. The path planner is generating the desired path, while in the sequel this path (planar coordinates) is translated to displacement, heading and curvature coordinates that act as the reference input for the MPC.

The formulation of the MPC is based on: a) the current full state feedback, b) the active constrains on the system, c) the estimated slip angles, d) the measured velocity and e) the mode selector signal, which depends on the slipping and the velocity of the vehicle, in order to apply the necessary switching optimal control action. The full derivation of the switching MPC is shown in [*PaperB*].

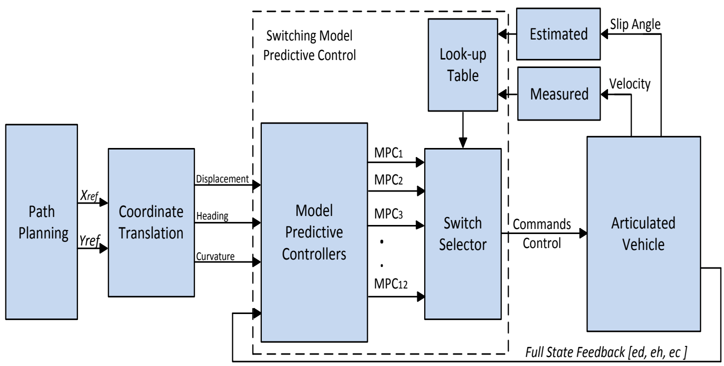

Figure 2.4: Switching model predictive controller scheme block diagram for an articulated vehicle

2.5 Path Planning Techniques

In this thesis, a simple and efficient navigation approach (Bug-like algorithm) for autonomous mobile vehicle has been proposed with the ability of avoiding obstacles, while converging to its final goal. In the proposed path planning scheme it is also assumed that the vehicle is able to online sense the surrounding environment based on the available sensory systems.

2.5.1 Path Planning Algorithm

As in all the path planning algorithms, the problem addressed in this section is being related to the problem of a vehicle moving from a starting point to the goal point, while detecting and avoiding identified obstacles based on the real vehicles kinematic equations of motion. As a common property of Bug-like algorithms, the proposed scheme initially faces the vehicle towards the goal point, which is assumed to be a priori known. In the proposed path planning module it is also assumed that the vehicle is able to online sense the surrounding environment based on the available sensory systems.

The proposed scheme is able to replan the produced path, by generating new way points, after the identification of an obstacle and produce proper path deformations that need to be done for avoiding it. In all these cases the produced set of new way points are utilized as control points. The algorithm has an ability to tune online the articulated steering angle while converging to the goal point.

The assumed sensory system is able to detect the obstacles and the surrounding environment, measure the distance of the articulated robot from the obstacle d_{obs}, while a sensing radius θ_{obs} is being considered, reflecting real life sensing limitations. The obstacle avoidance strategy becomes active when the safety conditions d_{min} and θ_{min} are violated. The overall flowchart diagram for the proposed path planning and obstacle avoidance scheme for the case of an articulated vehicle is being depicted in Figure 2.5, and can be summarized as follow:

[Step 1: Initialization] Define initial $[X_r\ Y_r\ \theta_r\ \dot{\gamma}_r]$ and goal $[X_g\ Y_g]$, define the articulated vehicle's

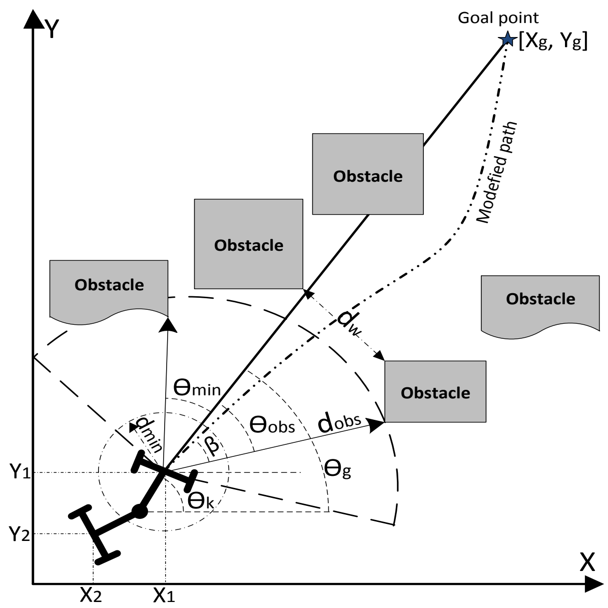

Figure 2.5: Notation and overall concept of the proposed path planning algorithm.

specific parameters V, d_{min}, θ_{min}, d_{obs}, θ_{obs}, the path update rate defined as T and the vehicle's mechanical and physical constraints that needs to be taken under consideration. Set $[X_k\ Y_k\ \theta_k\ ] = [X_r\ Y_r\ \theta_r]$, with $k \in Z^+$ the sample index.

[Step 2: Path Update] Update the coordinates of the next way point as:

$$\begin{bmatrix} X_{k+1} \\ Y_{k+1} \\ \theta_{k+1} \end{bmatrix} = \begin{bmatrix} X_k + v_k \cos(\theta_k + \gamma_k) \\ Y_k + v_k \sin(\theta_k + \gamma_k) \\ \theta_k + \frac{v_k \sin\gamma + l_2 \dot{\gamma}}{l_1 \cos\gamma + l_2} \end{bmatrix} \tag{2.24}$$

to produce γ with θ_g the angle between the line that connects the center of gravity of the vehicle's front part to the goal point and the X axis, while β is the difference angle among the vehicle's orientation angle, with the X axis and θ_g. During the application of this step, constraints can be imposed on the articulated vehicle just by bounding the allowable articulation within $\gamma^+ \leq \dot{\gamma} \leq \gamma^-$, with $^+$ and $^-$ representing the maximum and the minimum bounds on the rate of articulated angle.

[Step 3: Obstacle Avoidance] The obstacle avoidance strategy becomes active when the safety conditions d_{min} and θ_{min} are violated. This can be evaluated by calculating the update of the distance from the obstacle and the obstacle's angle by:

$$\begin{aligned} dis_{obs} &= \sqrt{(X_{k+1} - X_{obs})^2 + (Y_{k+1} - Y_{obs})^2} \\ \theta_{obs} &= \tan^{-1} \frac{Y_{k+1} - Y_{obs}}{X_{k+1} - X_{obs}} \\ \theta_{obs,g} &= \theta_{obs} - \theta_g \end{aligned}$$

while in the case that the following conditions are true:

$$\begin{gathered} (D_{obs} < d_{min}) \quad and \quad (\theta_{obs,g} < \theta_{min}) \\ OR \\ (D_{obs} < d_{min}) \quad and \quad (\theta_{obs,g} < -\theta_{min}) \end{gathered}$$

the changing in the steering angle is being duplicated and **Step3** is repeated again till the conditions in **Step3** is false and the algorithm continues from **Step2** or the bounds on the articulated angle cannot meet and the algorithm jumps to **Step4**.

[Step 4: Reaching Final Goal] If $[X_{k+1} \ Y_{k+1}] = [X_g \ Y_g] \pm D_{tolr}$, set $velocity = 0$ and the path planning algorithm has been terminated. Otherwise, the algorithm jumps to **Step2** and the whole process is being repeated in order to avoid collisions with the obstacles, until vehicle reaches the goal tolerance distance.

During the execution of the proposed algorithm, and especially **Step2**, the proposed path planning scheme always smoothes the produced way points by the utilization of Bezier curve filtering. The proposed scheme is able to replan the produced path, by generating new way points, after the identification of an obstacle and produce proper path deformations that need to be done for avoiding it.

It should be noted that in the presented approach all the obstacles and the surrounding environment are being considered as point clouds in a 2–dimensional space. Overlapping obstacles are being merged and represented by a single and unified obstacle. The notations utilized and the overall concept of the proposed path planning algorithm are depicted in Figure 2.6. The full derivation of the proposed path planning scheme is shown in [*PaperD*]

2.6 On Line Path Planning with A Controller

The proposed algorithm in this research belongs to the case of on line path planning algorithm for an articulated vehicle, moving in a partially known and sensory based reconstructed environment, and

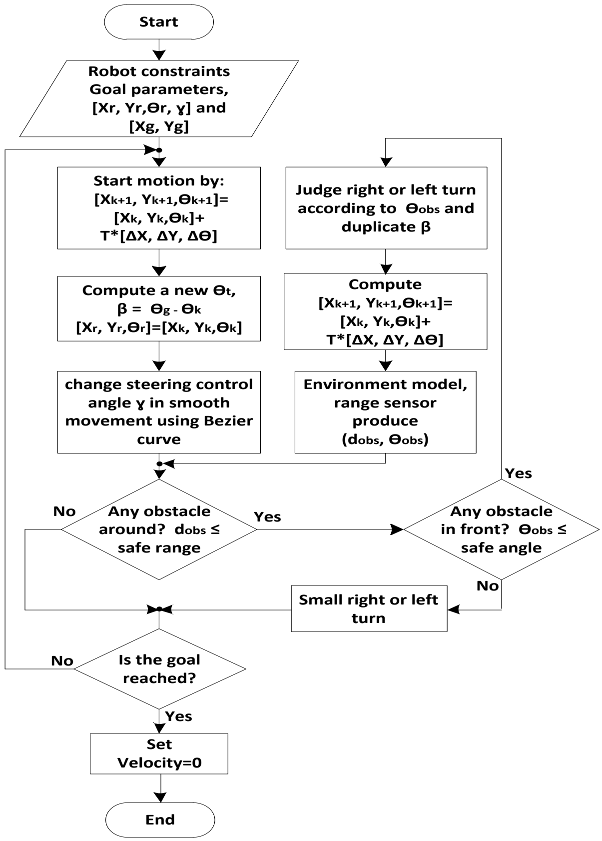

Figure 2.6: Main flowchart of path planning motion

relying on MPC. The algorithm is able to tune online the articulated steering angle in order to drive the front and the rear parts of the vehicle from avoiding collision with obstacles, while converging to the goal point. The proposed path planning algorithm is able to produce on–line the next reference way point, solving the local and sub-optimal problem, while in the sequel MPC has been utilized for creating the proper control signal, which is the rate of the articulated angle based on an error dynamics kinematic model of the vehicle.

Before feeding the MPC with the reference coordinates, a proper conversion is taking place from the cartesian space to the curvature, heading and distance space, the states of the vehicle's error dynamic states. The overall proposed concept of path planning and MPC is depicted in Figure 2.7. As it can be observed from this diagram the algorithm starts by defining the current position and orientation of the vehicle, denoted by $[X_r, Y_r, \theta_r]$ and the final goal position denoted by $[X_g, Y_g, \theta_g]$. Based on the onboard sensory system, the vehicle identifies the surrounding environment and obstacles and generates on–line the way points for reaching the goal destination by solving the local and by definition sub–optimal path planning problem.

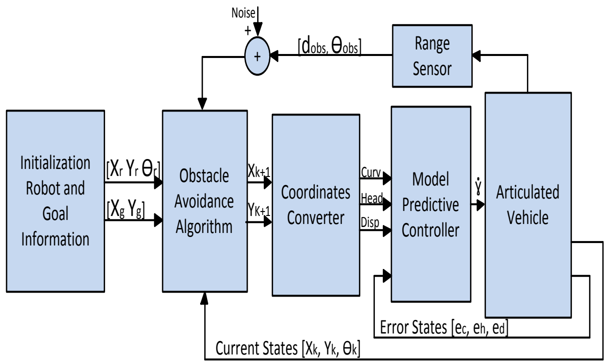

Figure 2.7: The combined on–line path planning and model predictive control for an articulated vehicle.

In the sequel the way points are translated to references for the error dynamics based MP controller. The control scheme generates the rate of articulated angle, which is the control signal for the vehicle. In the presented architecture it is assumed that accurate and continuous position and orientation measurements as well as corresponding updates are provided, without any loss of generality.

CHAPTER 3

Contributions and Future Work

Appendix Papers

- **Paper A**: T. Nayl, G. Nikolakopoulos, and T. Gustafsson, "*Kinematic modeling and extended simulation studies of a load hull dumping vehicle under the presence of slip angles*", Proceedings of IASTED International Conference on Modelling, Simulation and Identification, Pittsburgh, USA, ACTA Press ,pp.344-349, November 2011.

- **Paper B**: T. Nayl, G. Nikolakopoulos, and T. Gustafsson, "*Switching model predictive control for an articulated vehicle under varying slip angle*", Proceedings of 20th IEEE Mediterranean Conference on Control and Automation, (MED 2012), pp.884-889, July 2012.

- **Paper C**: T. Nayl, G. Nikolakopoulos, and T. Gustafsson, "*Path following for an articulated vehicle based on switching model predictive control under varying speeds and slip angles*", Proceedings of 17th IEEE International Conference on Emerging Technology Factory Automation, (ETFA 2012), Krakow, Poland, September, 2012.

- **Paper D**: T. Nayl, G. Nikolakopoulos, and T. Gustafsson, "*On line Dynamic Smooth Path Planning for an Articulated Vehicle*", 10th International Conference on Informatics in Control, Automation and Robotics (ICINCO 2013), Reykjavik, Iceland, July, 2013.

3.1 Contributions

The overall contribution of the licentiate have been analytically presented in four articles, attached in Part II. These research efforts contributed in the fields of: a) error modelling of articulated vehicle under the presence of slip angles, b) model predictive control schemes for switching error dynamics models for articulated vehicles, and c) path planning in off and online formulations, where the actual articulated dynamics have been considered. Regarding the articles in Part II the following contributions have been made:

- **Paper A: Kinematic modeling and extended simulation studies of a load hull dumping vehicle under the presence of slip angles**
  A kinematic model for an articulated vehicle, with the ability to take under consideration the slipping angles has been derived. Exhaustive simulation studies has been provided for evaluating the performance of the model under multiple design and operational variables, such as: a) multiple parameters sets for slip angles, b) different steering angles, c) various wagons' lengths, and d) various driving paths.

- **Paper B: Switching Model Predictive Control for an Articulated Vehicle Under Varying Slip Angle**
  A switching model predictive control scheme for an articulated vehicle under varying slip angles has been presented. For the articulated vehicle, the non–linear kinematic model that is able to take under consideration the effect of the slip angles was extracted. This model was transformed into an error dynamics model, which in the sequence was linearized around multiple nominal slip angle cases. The existence of the slip angles had a significant effect on the vehicle's path tracking capability and can significantly deteriorate the performance of the overall control scheme. Based on the derived multiple error dynamic models, the varying slip angle has been considered as the switching rule and a corresponding switching mode predictive control scheme was designed that it was also able to take under consideration: a) the constrains on the control signals and b) the state constraints.

- **Paper C: Path Following for an Articulated Vehicle Based on Switching Model Predictive Control Under Different Speeds and Slip Angles**
  This article focused on the problem of path following for an articulated vehicle under different velocities and slip conditions. The proposed control architecture consisted of a switching control scheme based on multiple model predictive controllers, fine tuned for dealing with different operating speeds and slip angles. The existence of the slipping and varying speed had a significant effect on the vehicle's path following capability and could significantly deteriorate the performance of the overall control scheme. Based on the derived multiple dynamics modeling, the current slip and vehicle's speed were considered as the signal selector for the proposed switching model predictive control scheme.

- **Paper D: On line Dynamic Smooth Path Planning for an Articulated Vehicle**
  On line dynamic smooth path planning scheme based on a bug-like modified path planning algorithm for an articulated vehicle under limited and sensory reconstructed surrounding static

environment has been presented. In the general case, collision avoidance techniques can be performed by altering the articulated steering angle to drive the front and rear parts of the articulated vehicle away from the obstacles. In the presented approach factors such as the real dynamics of the articulated vehicle, the initial and the goal configuration (displacement and orientation), minimum and total travel distance between the current and the goal points, and the geometry of the operational space were taken under consideration to calculate the update on the future way points for the articulated vehicle. In the sequel the produced path planning is online and iteratively smoothed by the using of Bèzier lines before producing the necessary rate of change for the vehicle's articulated angle.

3.2 Future Work

The main aim of the future activities will be to experimentally evaluate the proposed modeling control and path planning schemes in a small scale platform and sequentially on the VOLVO LHD vehicle that the control engineering group has. The activities concerning the future work can be summarized in the following objectives:

- **Experimental Tuning**
  Experimental tuning of the proposed theory in real life application to validate the proposed algorithms on a small scale articulated vehicle.

- **Dynamic Modelling**
  The future work will include the effect of dynamic coefficients of the vehicle model, which includes mass, tires, friction.

- **Parameters Estimation**
  Estimate the vehicle position by knowing the vehicle orientation and velocity using dead reckoning method, which is often utilized in mobile vehicle applications.

REFERENCES

[1] P. I. Corke and P. Ridley, "Steering kinematics for a center-articulated mobile robot," *Robotics and Automation, IEEE Transactions on*, vol. 17, no. 2, pp. 215–218, 2001.

[2] V. Polotski and A. Hemami, "Control of articulated vehicle for mining applications: Modeling and laboratory experiments," in *Control Applications, Proceedings IEEE International Conference on, Hartford, CT*, pp. 318–323.

[3] Y. Yavin, "Modelling the motion of an underground mining vehicle," *Mathematical and computer modelling*, vol. 42, no. 9, pp. 1123–1130, 2005.

[4] B. Dragt, F. Camisani-Calzolari, and I. Craig, "Modelling the dynamics of a loadhaul-dump vehicle," in *Proceedings of the 16th IFAC World Congress, Czech Republic*, vol. 16, no. 1, 2005.

[5] F. Kühne, W. F. Lages, and J. GOMES da SILVA, "Mobile robot trajectory tracking using model predictive control," in *II IEEE Latin-American Robotics Symposium, Brazil*, 2005.

[6] B. P. Gerkey and M. J. Matarić, "Principled communication for dynamic multi-robot task allocation," in *Experimental Robotics VII, Berlin*. Springer, 2001, vol. 271, pp. 353–362.

[7] E. Falcone, R. Gockley, E. Porter, and I. Nourbakhsh, "The personal rover project:: The comprehensive design of a domestic personal robot," *Robotics and Autonomous Systems*, vol. 42, no. 3, pp. 245–258, 2003.

[8] K. Schilling and C. Jungius, "Mobile robots for planetary exploration," *Control Engineering Practice*, vol. 4, no. 4, pp. 513–524, 1996.

[9] P. Petrov and P. Bigras, "A practical approach to feedback path control for an articulated mining vehicle," in *Intelligent Robots and Systems, 2001. Proceedings. 2001 IEEE/RSJ International Conference on*, vol. 4. IEEE, 2001, pp. 2258–2263.

[10] P. I. Corke and P. Ridley, "Steering kinematics for a center articulated mobile robot," *IEEE Trans. On Robotics and Automation*, vol. 17, no. 2, pp. 215–218, 2001.

[11] R. DeSantis, "Modeling and path-tracking for a load-haul-dump mining vehicle," *Journal of Dynamic Systems, Measurement, and Control*, vol. 119, no. 1, pp. 40–47, 1997.

[12] A.-M. Zou, Z.-G. Hou, M. Tan, X.-J. Chen, and Y.-C. Zhang, "Tracking control of a mobile robot with kinematic uncertainty using neural networks," in *Neural Information Processing*, vol. 4234. Springer, 2006, pp. 721–730.

[13] C. Altafini, “A path-tracking criterion for an lhd articulated vehicle,” *The International Journal of Robotics Research*, vol. 18, no. 5, pp. 435–441, 1999.

[14] B. T. Thumati, T. Dierks, and J. Sarangapani, “A model-based fault tolerant control design for nonholonomic mobile robots in formation,” *The Journal of Defense Modeling and Simulation: Applications, Methodology, Technology*, vol. 9, no. 1, pp. 17–31, 2012.

[15] T. Liu and Z.-P. Jiang, “Distributed formation control of nonholonomic mobile robots without global position measurements,” *Automatica*, vol. 49, no. 2, pp. 592–600, 2012.

[16] A. Ailon and I. Zohar, “Control strategies for driving a group of nonholonomic kinematic mobile robots in formation along a time-parameterized path,” *Mechatronics, IEEE/ASME Transactions on*, vol. 17, no. 2, pp. 326–336, 2012.

[17] R. Pepy, A. Lambert, and H. Mounier, “Path planning using a dynamic vehicle model,” in *Information and Communication Technologies, ICTTA’06. 2nd*, vol. 1. IEEE, 2006, pp. 781–786.

[18] C. Altafini, “Why to use an articulated vehicle in underground mining operations?” in *Robotics and Automation. Proceedings IEEE International Conference on*, vol. 4. IEEE, 1999, pp. 3020–3025.

[19] P. Falcone, F. Borrelli, J. Asgari, H. E. Tseng, and D. Hrovat, “Predictive active steering control for autonomous vehicle systems,” *Control Systems Technology, IEEE Transactions on*, vol. 15, no. 3, pp. 566–580, 2007.

[20] P. F. Muir and C. P. Neuman, “Kinematic modeling for feedback control of an omnidirectional wheeled mobile robot,” in *Autonomous robot vehicles*. Springer, 1990, vol. 1, pp. 25–31.

[21] W. L. Nelson and I. J. Cox, “Local path control for an autonomous vehicle,” in *Autonomous robot vehicles*. Springer, 1990, vol. 1, pp. 38–44.

[22] Y. Kanayama, Y. Kimura, F. Miyazaki, and T. Noguchi, “A stable tracking control method for an autonomous mobile robot,” in *Robotics and Automation, Proceedings IEEE International Conference on, Cincinnati, OH*, vol. 1. IEEE, 1990, pp. 384–389.

[23] C. Samson and K. Ait-Abderrahim, “Feedback control of a nonholonomic wheeled cart in cartesian space,” in *Robotics and Automation, Proceedings, IEEE International Conference on, Sacramento, CA*. IEEE, 1991, pp. 1136–1141.

[24] O. Hachour, “Path planning of autonomous mobile robot,” *International Journal of Systems Applications, Engineering & Development*, vol. 2, no. 4, pp. 178–190, 2008.

[25] J. Gutiérrez, D. Apostolopoulos, and J. L. Gordillo, “Numerical comparison of steering geometries for robotic vehicles by modeling positioning error,” *Autonomous Robots*, vol. 23, no. 2, pp. 147–159, 2007.

[26] J. Markdahl, G. Bortolin, and U. Andersson, “Traction control for articulated off-road vehicles,” Ph.D. dissertation, Master Thesis, 2010.

[27] U. Larsson, C. Zell, K. Hyyppa, and A. Wernersson, “Navigating an articulated vehicle and reversing with a trailer,” in *Robotics and Automation, Proceedings, IEEE International Conference on, San Diego*. IEEE, 1994, pp. 2398–2404.

[28] Y. Wang, B. M. Nguyen, P. Kotchapansompote, H. Fujimoto, and Y. Hori, “Vision-based vehicle body slip angle estimation with multi-rate kalman filter considering time delay,” in *Industrial Electronics (ISIE), 2012 IEEE International Symposium on*. IEEE, 2012, pp. 1506–1511.

[29] J. Barraquand and J.-C. Latombe, “On nonholonomic mobile robots and optimal maneuvering,” in *Intelligent Control, Proceedings, IEEE International Symposium on, Albany*. IEEE, 1989, pp. 340–347.

[30] N. Buniyamin, N. Sariff, W. Wan Ngah, and Z. Mohamad, “Robot global path planning overview and a variation of ant colony system algorithm,” *International Journal of Mathematics and Computers in Simulation*, vol. 5, no. 1, pp. 9–16, 2011.

[31] S. Scheding, G. issanayake, E. Nebot, and H. Durrant, “Slip Modeling and Aided Inertial Navigation of an LHD,” in *International Conference on Robotics and Automation*, New Mexico, 1997.

[32] P. Ridley and P. Corke, “Load haul dump vehicle kinematic and control,” in *Transaction of the ASME*, 2003, pp. 54–59.

[33] F. Núnez, S. Navarro, A. Aguado, and A. Cipriano, “State estimation based model predictive control for lhd vehicle,” in *17th World Congress, the International Federation of Automatic Control, Seoul, Korea*, 2008.

Part II

Paper A

Modeling and Control of an Autonomous Articulated Vehicle

Authors:
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Modeling and Control of an Autonomous Articulated Vehicle

Thaker Nayl, George Nikolakopoulos and Thomas Gustafsson

Abstract: In this article a kinematic model for a Load Haul Dumping Vehicle (LHD), with the ability to take under consideration the slipping angles is being derived. This specific type of vehicles consists of two parts, a steering tractor and a trailer, linked with rigid free joint and can be found various applications, especially in the mine fields. In the rest of the article, exhaustive simulation studies are being provided for evaluating the performance of the model under multiple design and operational variables, such as: a) multiple parameters sets for slip angles, b) different steering angles, c) various wagons' lengths, and d) various driving paths.

1 Introduction

Recently, there has been a significant advance in designing automated vehicles for their utilization in the mining industry, with a general aim to improve the mining methods and increase the technological readiness in autonomous operations [1].

Load Haul Dump mining vehicles, as the one depicted in Figure 1, belong to the previous category of processes, that the relative current research is creating a roadmap for their complete autonomy. In this roadmap, the initial aim is the full modelling of LHD vehicles and the conduction of a variety of simulations for: a) evaluating the accuracy of the provided LHDs' models and b) producing knowledge that would assist in the initial process of designing such vehicles.

Figure 1: Typical Load Haul Dump Vehicle

Several kinematic models have been recently proposed in the literatures [2–5], for articulated tracks with two or multiple body dynamics. Most of these methods contain simple models that are not taking under consideration the effect of the slipping angles, as the problem is being complicated and it is more difficult to proceed to the next stage of control development [6, 7]. Moreover, there has been a variety of recent articles, focusing only on the calculation of sleeping angles, and the way that the external environment is affecting them [8], or examining the overall problem of kinematic modelling of a multi-body vehicle (multiple trailers), with articulated joints [9].

The novelty of this article stems from: a) the presentation of full kinematic models (with and without slip angles) for the specific case of LHD vehicles, b) the comparison of the overall complexity from two models, under the same configurations and in the same driving paths, and c) the evaluation of the provided model's performance under multiple design and operational variables, such as: a) multiple parameters sets for slip angles, b) different steering angles, c) various wagons' lengths, and d) various driving paths.

The rest of the article is structured as it follows. In Section 2, the Load Haul Dumping Vehicle modelling is presented for the cases of with and without slipping angles. In Section 3, the evaluation criteria and remarks on the simulation strategy adopted are being presented, while in Section 4, extended simulation results are being presented. Finally, in Section 5, the conclusions are drawn.

2 Load Haul Dumping Vehicle Modeling

LHD vehicles belong to the category of articulated vehicles that consist of two parts, a tractor and a trailer, linked with rigid free joint [10]. Each body has a single axle, and the wheels are all non-steerable, the steering action is performed on the joint, by changing the corresponding articulated angle, between the front and rear parts of the vehicle. In the following presentation, it is being assumed that the two wheels of each axle can independently rotate and that the steering action on the joint is coordinated with a differential drive on the actuated wheels;

There are various external factors that degrade the system performance and introduce errors in the model that propagate with time. These errors that could also be considered as external disturbances in the system model, are being caused by many factors, such as: a) the generic interaction between the vehicle and its environment [11], b) the noise and bias in positioning and driving sensors [12], c) existence or variation of slipping angles among the trailer and the tractor depending on the type of the driving ground, the fatigue and type of the tyres [13], d) and dynamic effects resulting from acceleration and braking [14, 15].

In the following subsections, the initial kinematic model of an LHD, without the slip angles will be presented (Section 2.1.), followed by a full kinematic model, where different slip angles, for the trailer and the tractor, have been considered (Section 2.2).

2.1 LHD Model without Slip Angle

The case of an LHD without slip angles is being presented in Figure 2.

where (x_f, y_f) and (x_r, y_r) denote the positions of the front and the rear part of the vehicle respectively, or the corresponding coordinates of the P_f and the P_r centres of gravity, L_f and L_r are the length of the front and the rear units, the angles θ_f and θ_r denote the vehicle's orientation for the front and rear parts, with respect to the fixed coordinated system, denoted by the x and y axes. Moreover v_f is the linear velocity of the tractor, v_r is the linear velocity of the trailer, and γ is the articulated angle. As it has been depicted in Figure 2, the motion of the vehicle's front part (tractor) can be defined as:

$$\dot{x}_f = v_f \; cos \; \theta_f \tag{1}$$

$$\dot{y}_f = v_f \; sin \; \theta_f \tag{2}$$

By examining the geometry of the vehicle, and the relation between the coordinates of P_f and P_r , the equations of motion for the LHD's rear part (trailer), can be derived from the following equations (3) and (4):

$$x_r = x_f - L_f cos\theta_f - L_r cos\theta_r \tag{3}$$

$$y_r = y_f - L_f sin\theta_f - L_r sin\theta_r \tag{4}$$

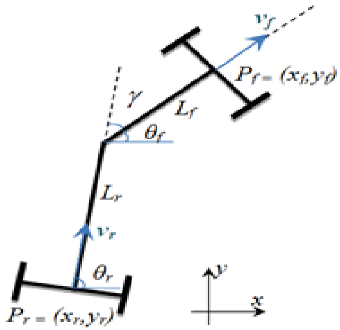

Figure 2: LHD modelling configuration without slip angles.

While the articulated angle γ, is being defined as the difference between the orientation angle θ_r of the tractor and the orientation angle θ_f of the trailer:

$$\gamma = \theta_f - \theta_r \tag{5}$$

In case that the nonholonomic constraints, acting on the front and rear axles, are taken under consideration, mainly resulting from the assumption of rolling without slipping for the wheels of the vehicle, the following equations (6) and (7) can also be extracted:

$$\dot{x}_f sin\theta_f - \dot{y}_f cos\theta_f = 0 \tag{6}$$
$$\dot{x}_r sin\theta_r - \dot{y}_r cos\theta_r = 0 \tag{7}$$

By taking the derivative of the equations (3), (4) with respect to time, substituting the results in equations (6), and (7), and simplifying the resulting ones yields:

$$\dot{\theta}_f = \frac{v_f\ sin\gamma + L_r\ \dot{\gamma}}{L_f\ cos\gamma + L_r} \tag{8}$$

Where the rate of change for the articulated angle$\dot{\gamma}$ can be defined by the utilization of (5) as:

$$\dot{\theta}_r = \dot{\theta}_f - \dot{\gamma} \tag{9}$$

By solving equations (8) and (9) with respect to θ_r, the following dynamic equation can be obtained as:

$$\dot{\theta}_r = \frac{v_f\ sin\gamma - L_f\ \dot{\gamma}\ cos\gamma}{L_f\ cos\gamma + L_r} \tag{10}$$

From (8) and (9), it is clear that the angular velocities θ_f and θ_r have different values when $L_f \neq L_r$, or the vehicle is not driving straight, or $\gamma \neq 0$. By that a slippage angle is introduced, when γ is increased, and thus the general performance of the model is deteriorated. Moreover, v_f can be calculated from the geometry of the vehicle; the relation between the two speed vectors v_f and v_r is provided by:

$$v_r = v_f\ cos\gamma\ +\ L_f\ \dot{\theta}_f\ sin\gamma \tag{11}$$

Substitution of equation (11) in (8) and by simplification, the dependence of f v from r v can be defined as

$$v_r = \frac{-v_f\, L_f + v_f\, L_r\, cos\gamma - L_f\, L_r\, \dot{\gamma}\, sin\gamma}{L_f\, cos\gamma + L_r} \tag{12}$$

2.2 LHD model including slip angles

A kinematic model for the LHD vehicle can be formulated from general geometry that includes ideal and slip behaviour for the steering angles. The definition of this model has been initially based on the derivation in [8], which included the perturbed factors in the vehicle position as two slip variables. Those variables are the rear and front slip angles, α and β, which are also the angles between the linear velocities of the vehicle and the rear trailer and the front tractor. For the following derivation of the corresponding LHD kinematic model, it is being assumed that the vehicle is moving on a flat surface under the influence of slip angles. In the case under study, the vehicle configuration is depicted in Figure 3, where all the presented variables, except from the slip angles, have been defined in the previous subsection 2.1, while the subscript ‘s’ denotes the variable in the slip angle case. The

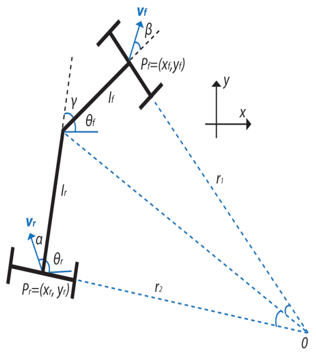

Figure 3: LHD modeling configuration under the influence of slip angles.

kinematic equations, for the motion under the effect of slip angles, for the P_{rs} point can be formulated as:

$$\dot{x}_{rs} = v_r\, cos(\alpha + \theta_r) \tag{13}$$
$$\dot{y}_{rs} = v_r\, sin)\alpha + \theta_r) \tag{14}$$

In the examined case, it can be observed that the vehicle motion depends not only on vehicle’s speeds v_r, the articulated angle γ and the vehicle’s lengths L_f and L_r, but also on the slip angles α and β, while the resulting vehicle’s heading is provided by a combination of the slip angles with the orientation angles θ_{fs} and θ_{rs}, as it has been presented in Figure 3. The rate of the orientation $\dot{\theta}_{rs}$ is a function of the steering angle γ and both slip angles α and β. Based on the assumption that the vehicle develops a steady-state motion turning $\dot{\gamma} \neq 0$, this rate can be provided by the utilization of a virtual centre of rotation, depending on the velocity as a function of $(cos\beta, sin\beta)$. With respect to the local coordinate axis origin, the velocities v_{fs} and v_{rs} can be defined as:

$$v_{fs} = \begin{bmatrix} cos\beta & 0 \\ sin\beta & -L_f \end{bmatrix} \begin{bmatrix} v_{fs} \\ \dot{\theta}_{fs} \end{bmatrix} \tag{15}$$

$$v_{rs} = \begin{bmatrix} cos\alpha & 0 \\ sin\alpha & L_r \end{bmatrix} \begin{bmatrix} v_{rs} \\ \dot{\theta}_{rs} \end{bmatrix} \tag{16}$$

The velocity of the trailer with respect to the velocity of the tractor, by utilizing rigid body dynamic principles, can be calculated, with respect to the articulated angle γ as it follows:

$$v_{rs} = \begin{bmatrix} cos\alpha & 0 \\ sin\alpha & L_r \end{bmatrix}^{-1} \begin{bmatrix} cos\gamma & -sin\gamma \\ sin\gamma & cos\gamma \end{bmatrix} \begin{bmatrix} cos\beta & 0 \\ sin\beta & -L_f \end{bmatrix} \begin{bmatrix} v_{fs} \\ \dot{\theta} fs \end{bmatrix} \tag{17}$$

By utilizing the equations in (15) and (16) in (17), the angular speed for the tractor, with respect to: a) the slip angles, b) the articulated angle, and c) the lengths of the tractor and trailer, can be defined as:

$$\dot{\theta}_{fs} = \frac{v_f \, sin(\gamma + \beta - \alpha) + L_r \, \dot{\gamma} \, cos\alpha}{L_f \, cos(\gamma - \alpha) + L_r \, cos\alpha} \tag{18}$$

It should be noted that equation (18) is more accurate if the slip angels are known a priori, a task that is quite difficult as the slip angles are dependent of the vehicle speed, mass, tyre-terrain interaction and articulation angle, in a highly non-linear way. Moreover, in calculating the angular velocity of the tractor, the primary sources of the errors, are due to the time varying parameters γ, $\dot{\gamma}$, β and α , as errors in these parameters propagate directly through to the states. The variables γ and $\dot{\gamma}$ can be measured with a great accuracy, while an estimation algorithm or a look up table should be utilized for the slip parameters.

3 Simulation Methodology

The performance of the proposed models will be evaluated under multiple design and operational variables, such as: a) multiple parameters sets for slip angles, b) different steering angles, c) various wagons' lengths, and d) various driving paths. This performance evaluation will be performed based on standard criteria, such as: a) the method of the normalized Positioning Error Ratio (PER), and b) the Mean Squared Error (MSE) indices. In general the rate of change of a wagon orientation is being defined as a function of the steering angle γ and both slip angles (α, β). Furthermore, based on the assumption that the vehicle develops a steady-state motion turning $(i.e. \dot{\gamma} = 0)$, the rate of orientation change is provided by:

$$\dot{\theta}_s = \frac{v_s}{R_s} \tag{19}$$

Where R_s is the turning radius, in the case of considering a movement under the effect of slip angles. Any change experienced in this radius, due to slip, leads to an error in position estimation. This effect of the slip angles on vehicle motion can be expressed as a normalized Positioning Error Ratio, which is the ratio of the slip radius against the ideal radius and can be defined as:

$$PER_{\theta} = 1 - \frac{\dot{\theta} s}{\dot{\theta}} \tag{20}$$

As an example, the PER for the turning angle of the tractor is being defined by the utilization of (8) and (18) with $\dot{\gamma} = 0$ and can be formulated as:

$$PER_{\theta} = 1 - \frac{sin\gamma(L_r\, cos\alpha + L_f\, cos(\gamma - \alpha))}{sin(\gamma + \beta - \alpha)(L_r + L_f cos\gamma)} \tag{21}$$

Finally, for evaluating the positioning errors among the actual path of the vehicle's wagons and the desired one, the classical Root Mean Square Error (RMSE) has been utilized, defined as:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(K_f(n) - K_r(n))}{n}} \tag{22}$$

Where n is the number of samples, and (k_f, k_r) representing the coordinates of the tractor and the trailer respectively.

4 Simulation Results

In the following simulation results, the speeds of the tractor and the trailer are constant in all the cases examined. In the first simulated test case, the effect of the slip angles on the path of the LHD has been considered, for two different paths (open loop), an S-curve and a spiral one. The results from these simulations are depicted in Figures 4 and 5 respectively. While in Figures 6 and 7, the MSEs

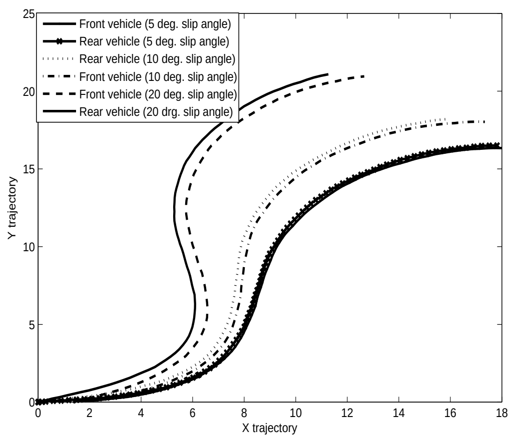

Figure 4: Effect of slip angles on the LHD's path for a S-curve.

for the tractor are being presented, for the trajectories in Figures 4 and 5 respectively. Similar results have been obtained when the trajectory of the trailer has been examined, for the same trajectories. From Figures 4 and 5, it can be observed that the slip angles introduce significant bias between the desired trajectory and the obtained ones. Without slip angles, there is a zero error among the reference trajectory and the obtained one. On the contrary, with slip angles, it has been observed that the bigger

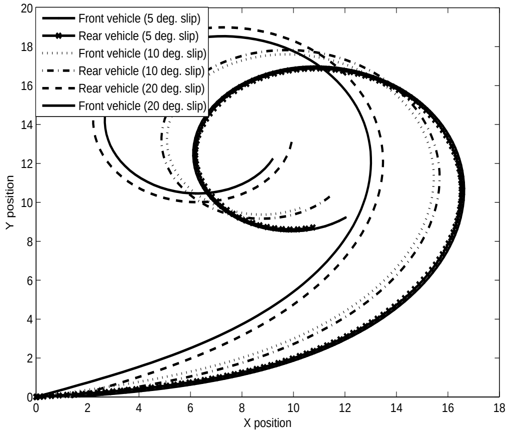

Figure 5: Effect of slip angles on the LHD's path for a Spiral-curve.

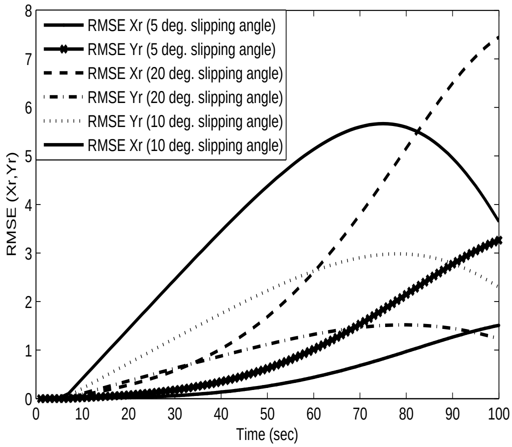

Figure 6: MSE for the tractor displacement under the effect of slip angles for a S-curve path.

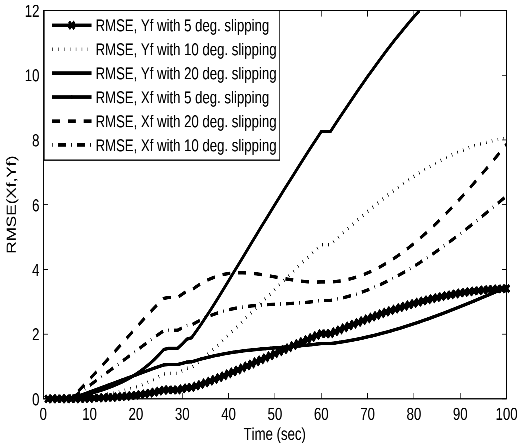

Figure 7: MSE for the tractor displacement under the effect of slip angles for a spiral-curve path.

Table 1: Simulation parameters for calculation PER with respect to the slip angles and various LHD lengths

Parameter	Values			
γ (rad)	1.047	1.047	1.047	1.047
β (rad)	0.087	0.174	0.349	0.523
α (rad)	0.087	0.174	0.349	0.523

the slip angle is, the bigger this deviation is and the application of a control scheme, able to count for this deviation, is mandatory for improving the driving performance of the LHD vehicle. In the following test cases, the PER indication has been considered. First, the articulated angles, and the LHD speed has been considered as constants, or $\gamma = 1.047rad$ and $v_r = 3m/sec$, while the PER has been calculated for different slip angles and for varying L_f and L_r lengths. More specifically, the values for this simulation have been the ones presented in Table 1, while the obtained results are depicted in Figure 8.

In the second examined test case for the PER graph, the articulated angle, the slip variables, and the speed of the LHD have been considered constant or $\gamma = 1.047rad$, $v_r = 3m/sec$, and $\alpha = \beta = 0.087$. For the examined test case, the PER graph is depicted in Figure 9, has been calculated for different vehicle lengths L_f and L_r, with different articulated angles, as presented in Table 2.

From the obtained results in Figures 8 and 9 it is obvious that the lengths of the tractor and the

Table 2: Simulation parameters for calculation PER with respect to the articulated angle and various LHD lengths

Parameter	Values			
γ (rad)	0.147	0.349	0.523	1.047

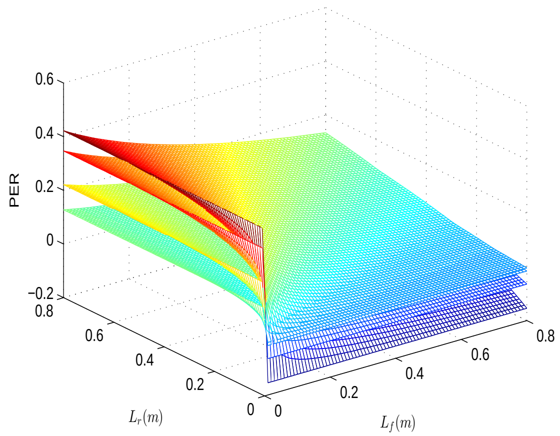

Figure 8: PER under constant articulated angle and speed and various slip angles and lengths.

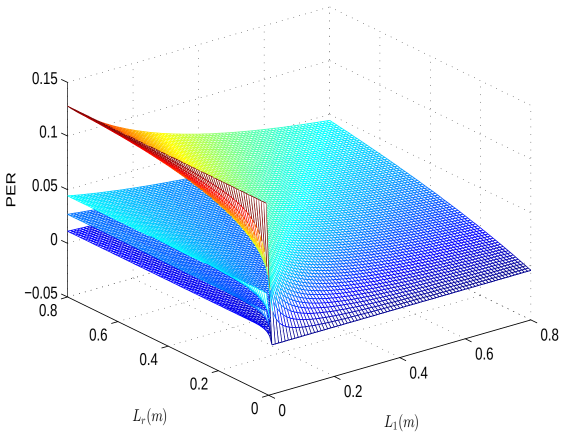

Figure 9: PER under constant articulated angle, LHD speed and slip angles as a function of the LHD's lengths.

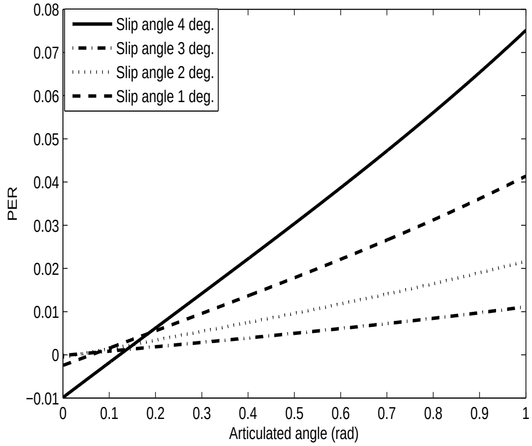

Figure 10: PER under constant lengths, and speed, varying articulated angle and varying constant slip angles.

trailer are playing a significant role in formulating the actual trajectory and as a general remark, the smaller the length of the tractor is, the smaller the effect of the slip angles and the articulated angle is. In the third simulated test case, the effect of a varying articulated angle $(0-\pi/3)$, under constant lengths $(L_f = 0.6m, L_r = 0.8m)$, and speed $(v_r = 3m/sec)$ and for various equal slip angles is being presented in Figure 10. From the results obtained in Figure 10, the effect of the slip angles and the selection of an articulated angle are depicted. For big articulated angles, the effect of the slip angles is evident, even if they have small values. This deviation is possible to lead to significant errors, especially in the cases where the slip angles have also a large value

5 Conclusions

In this article a kinematic model for a Load Haul Dumping Vehicle (LHD), with the ability to take under consideration the slipping angles has been derived. Extended simulation studies have been conducted for evaluating the performance of the model under multiple design and operational variables, such as: a) multiple parameters sets for slip angles, b) different steering angles, c) various wagons' lengths, and d) various driving paths.

References

[1] V. Deligiannis, G. Davrazos, S. Manesis, and T. Arampatzis, "Flatness conservation in the n-trailer system equipped with a sliding kingpin mechanism," *Journal of Intelligent and Robotic Systems*, vol. 46, no. 2, pp. 151–162, 2006.

[2] P. I. Corke and P. Ridley, "Steering kinematics for a center-articulated mobile robot," *Robotics and Automation, IEEE Transactions on*, vol. 17, no. 2, pp. 215–218, 2001.

[3] P. Petrov and P. Bigras, "A practical approach to feedback path control for an articulated mining vehicle," in *Intelligent Robots and Systems, 2001. Proceedings. 2001 IEEE/RSJ International Conference on*, vol. 4. IEEE, 2001, pp. 2258–2263.

[4] S. Manesis, N. Koussoulas, and G. Davrazos, "On the suppression of off-tracking in multi-articulated vehicles through a movable junction technique," *Journal of Intelligent and Robotic Systems*, vol. 37, no. 4, pp. 399–414, 2003.

[5] C. Altafini, "Why to use an articulated vehicle in underground mining operations?" in *Robotics and Automation. Proceedings IEEE International Conference on*, vol. 4. IEEE, 1999, pp. 3020–3025.

[6] S. Scheding, G. Dissanayake, E. M. Nebot, and H. Durrant-Whyte, "An experiment in autonomous navigation of an underground mining vehicle," *Robotics and Automation, IEEE Transactions on*, vol. 15, no. 1, pp. 85–95, 1999.

[7] R. DeSantis, "Modeling and path-tracking for a load-haul-dump mining vehicle," *Journal of Dynamic Systems, Measurement, and Control*, vol. 119, no. 1, pp. 40–47, 1997.

[8] S. Scheding, G. Dissanayake, E. Nebot, and H. Durrant-Whyte, "Slip modelling and aided inertial navigation of an lhd," in *Robotics and Automation, 1997. Proceedings., 1997 IEEE International Conference on*, vol. 3. IEEE, 1997, pp. 1904–1909.

[9] J. Steele, C. Ganesh, and A. Kleve, "Control and scale model simulation of sensor-guided lhd mining machines," *Industry Applications, IEEE Transactions on*, vol. 29, no. 6, pp. 1232–1238, 1993.

[10] J. Hannah and R. C. Stephens, *Mechanics of machines: elementary theory and examples*. Edward Arnold, 1984.

[11] J. Gutiérrez, D. Apostolopoulos, and J. L. Gordillo, "Numerical comparison of steering geometries for robotic vehicles by modeling positioning error," *Autonomous Robots*, vol. 23, no. 2, pp. 147–159, 2007.

[12] J. Markdahl, G. Bortolin, and U. Andersson, "Traction control for articulated off-road vehicles," Ph.D. dissertation, Master Thesis, 2010.

[13] U. Larsson, C. Zell, K. Hyyppa, and A. Wernersson, "Navigating an articulated vehicle and reversing with a trailer," in *Robotics and Automation, Proceedings, IEEE International Conference on, San Diego*. IEEE, 1994, pp. 2398–2404.

[14] C. Altafini, "A path-tracking criterion for an lhd articulated vehicle," *The International Journal of Robotics Research*, vol. 18, no. 5, pp. 435–441, 1999.

[15] P. Bolzern and A. Locatelli, "A comparative study of different solutions to the path-tracking problem for an articulated vehicle," in *Control Applications, 2002. Proceedings of the 2002 International Conference on*, vol. 1. IEEE, 2002, pp. 427–434.

Paper B

Switching Model Predictive Control for an Articulated Vehicle Under Varying Slip Angle

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Switching Model Predictive Control for an Articulated Vehicle Under Varying Slip Angle

Thaker Nayl, George Nikolakopoulos and Thomas Gustafsson

Abstract: In this article a switching model predictive control scheme for an articulated vehicle under varying slip angles is being presented. For the non–holonomic articulated vehicle, the non–linear kinematic model that is able to take under consideration the effect of the slip angles is extracted. This model is transformed into an error dynamics model, which in the sequence is linearized around multiple nominal slip angle cases. The existence of the slip angles has a significant effect on the vehicle's path tracking capability and can significantly deteriorate the performance of the overall control scheme. Based on the derived multiple error dynamic models, the varying slip angle is being considered as the switching rule and a corresponding switching mode predictive control scheme is being designed that it is also able to take under consideration: a) the constrains on the control signals and b) the state constraints. Multiple simulation results are being presented that prove the efficacy of the overall suggested scheme.

1 Introduction

Among the current vehicle types utilized in a mine field, articulated ones are the most characteristic vehicle's type that can be found most frequently, such as the Load Haul Dump (LHD) vehicles. In general the articulated vehicles consist of two parts, a tractor and a trailer, linked with a rigid free joint. Each body has a single axle and the wheels are all non–steerable, while the steering action is performed on the joint, by changing the corresponding articulated angle, between the front and the rear parts of the vehicle.

During the operation of an articulated vehicle, there are various external factors that degrade the overall system performance and introduce errors in the model that could be propagated with the time. Such deteriorating factors are: a) the generic interaction between the vehicle and its surrounding environment [1], b) the noise and bias in the positioning and driving sensors [2], c) the dynamic effects resulting from acceleration and braking [3], and d) the existence of variation of slip angles among the trailer and the tractor, which are also being influenced by the type of the driving ground, the fatigue and the tyres' type [4].

Among the aforementioned factors that degrade the overall performance of the articulated vehicle, the existence of slip angles is one of the most significant problem that the modeling and control approaches should face. In the relative literature there have been several research approaches for the problem of modeling articulated vehicles, based on the theory of multiple body dynamics [5–7]. Most of these methods contain simple models that are not taking under consideration the effect of the slip angles, as the complexity of the overall problem is being increased and it is more difficult to proceed to the next stage of the control scheme design based on highly non–linear articulated vehicle's dynamics.

From a control point of view, there have been proposed many traditional techniques for non–holonomic vehicles, based on error dynamics models without the presence of slip angles. More analytically, in [6,8] linear control feedback has been applied, while in [9] a Lyapunov based approach has been presented. In [10] a control scheme based on LMIs has been presented and in [11] a pole placement technique has been applied. Moreover, in [12] the authors have presented a path tracking controller based on error dynamics, while in [13] the problem of designing a path following controller for a n–trailer vehicle, based on non–linear adaptive control has been derived. Finally, classical fundamental problems of motion control for articulated vehicles have been presented in [7, 14, 15].

The main contribution of this article is dual. First, to the author's best knowledge, this is the first time that an error dynamics modeling framework will be derived for the case of an articulated vehicle operating under slip angles. Until now and mainly due to the increasing complexity of the problem, only the case of non–slip angles affecting the movement of the vehicle has been considered. Secondly, the problem of controling the articulated vehicle, under the presence of varying slip angles, is being addressed by a switching Model Predictive Control (MPC) scheme.
Multiple MPC controllers are being fine tuned for specific slippage operating conditions, while the proposed control scheme has the ability to take under consideration the effect of real life constraints on the control input (articulated angle) and the environmental restrictions. In this control design approach the current estimated slip angle of the tractor is considered as the mode selector for the switching MPC. The resulting control scheme provides the optimal control for each region of slip angles, while ensuring smooth transition of the control effort as the articulated vehicle is driven over regions of different slippage.

The rest of this article is structured as it follows. In Section 2 the error dynamics modeling framework for the cases of with and without the effect of the slip angles is being presented. In Section 3 the switching model predictive controller is being analyzed, while in Section 4 multiple simulation results are being presented that prove the efficacy of the proposed scheme. Finally the conclusions are draw in Section 5.

2 Articulated Vehicle Modeling

2.1 Articulated Vehicle Kinematics Without Slip Angles

The case of an articulated vehicle under no slip angles is being presented in Figure 1.

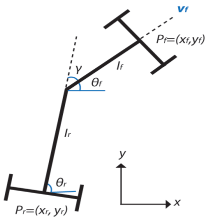

Figure 1: Typical articulated vehicle without slipping

In this figure (x_f, y_f) and (x_r, y_r) denote the coordinates of the front and the rear part of the vehicle, p_f and p_r are the corresponding centers of gravity, l_f and l_r are the length of the front and the rear units, while the angles θ_f and θ_r denote the vehicle's part orientation, and the x, y axes represent the fixed coordinating system. Moreover v_f and v_r are the linear velocities of the tractor and trailer respectively and are being defined as:

$$\dot{x}_f = v_f \cos\theta_f \quad (1)$$
$$\dot{y}_f = v_f \sin\theta_f \quad (2)$$

The articulated angle γ is being defined as the difference between the orientation angle θ_f of the front and the orientation angle θ_r of the rear part of the vehicle. By examining the vehicle's depicted geometry, as also the relation between the coordinates of p_f and p_r, the equations of motion for the vehicle's rear part, can be derived from the following equations:

$$x_r = x_f - l_f \cos\theta_f - l_r \cos\theta_r \quad (3)$$
$$y_r = y_f - l_f \sin\theta_f - l_r \sin\theta_r \quad (4)$$

When the non–holonomic constraints, acting on the front and rear axles, are taken under consideration, mainly resulting from the assumption of moving without slipping in the vehicle's wheels, the following equations (5–6) can also be extracted:

$$\dot{x}_f \, sin\theta_f - \dot{y}_f \, cos\theta_f = 0 \quad (5)$$
$$\dot{x}_r \, sin\theta_r - \dot{y}_r \, cos\theta_r = 0 \quad (6)$$

By taking the derivative of the equations (3–4) with respect to time, substituting the results in equations (5-6), and by further simplifying the results, the time derivative of the tractor's orientation angle is being defined as:

$$\dot{\theta}_f = \frac{v_f \, sin\gamma + l_r \, \dot{\gamma}}{l_f \, cos\gamma + l_r} \quad (7)$$

The angular velocities for the tractor and the trailer, which are being defined as $\dot{\theta}_f$ and $\dot{\theta}_r$ respectively, have different values when ($l_f \neq l_r$) or the vehicle is not driving straight ($\gamma \neq 0$).

$$\dot{\theta}_r = \frac{v_f \, sin\gamma - l_f \, \dot{\gamma} \, cos\gamma}{l_f \, cos\gamma + l_r} \quad (8)$$

2.2 Articulated Vehicle Kinematics Under Slip Angles

In the case of slip angles, the kinematic model of the articulated vehicle can be also formulated from general geometry that includes ideal and slip behavior for the steering angles. The definition of this model has been initially based on the derivation in [16], which included the perturbed factors in the vehicle position as two slip variables β and α, being defined as the tractor's and trailer's slip angles respectively, and which are also the angles between the linear velocities of the vehicle's parts.

For the following derivation of the kinematic model, it is being assumed that the vehicle is moving on a flat surface under the influence of slip angles, while the vehicle's configuration is depicted in Figure 2. where (r_1, r_2) are the instantaneous centers of velocity for the tractor and the trailer of the vehicle with different radiuses. For the initial center curvature of the trajectory it is being assumed that the vehicle is moving forward without slip conditions. In the following derivation, the unit subscript 's' denotes variables in the slip case as they have been defined previously for the non–slip one.

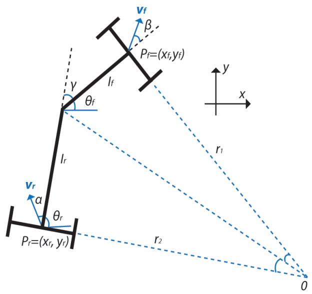

Figure 2: Articulated vehicle modeling configuration under the influence of slip angles

The kinematic equations, for the motion under the effect of slip angles, for the front part can be formulated as:

$$\dot{x}_{fs} = v_f \cos(\theta_f + \beta) \tag{9}$$

$$\dot{y}_{fs} = v_f \sin(\theta_f + \beta) \tag{10}$$

In this case, the vehicle motion depends not only on vehicle's speeds, the articulated angle and the vehicle's lengths, but also on the slip angles, while the resulting vehicle's heading is provided by a combination of the slip angles with the orientation angles. The rate of the orientation $\dot{\theta}_{rs}$ can be defined as function of the steering angle γ and both slip angles. Based on the assumption that the vehicle develops a steady–state motion turning, this rate can be provided by the utilization of a virtual center of rotation, depending on the velocity as a function of $(cos(\beta), sin(\beta))$ [2]. With respect to the local coordinate axis origin, the velocities are being defined as:

$$v_{fs} = \begin{bmatrix} cos\beta & 0 \\ sin\beta & -l_f \end{bmatrix} \begin{bmatrix} v_{fs} \\ \dot{\theta}fs \end{bmatrix} \tag{11}$$

$$v_{rs} = \begin{bmatrix} cos\alpha & 0 \\ sin\alpha & l_r \end{bmatrix} \begin{bmatrix} v_{rs} \\ \dot{\theta}rs \end{bmatrix} \tag{12}$$

By utilizing rigid body principles, the velocity of the tractor, with respect to the velocity of the trailer, can be calculated with respect to the articulated angle as it follows [17]:

$$v_{rs} = \begin{bmatrix} cos\alpha & 0 \\ sin\alpha & l_r \end{bmatrix}^{-1} \begin{bmatrix} cos\gamma & -sin\gamma \\ sin\gamma & cos\gamma \end{bmatrix} \begin{bmatrix} cos\beta & 0 \\ sin\beta & -l_r \end{bmatrix} \begin{bmatrix} v_{fs} \\ \dot{\theta}fs \end{bmatrix} \tag{13}$$

By replacing equations (11-12) in (13), the angular speed for the tractor can be defined as:

$$\dot{\theta}_{fs} = \frac{v_f \, sin(\gamma + \beta - \alpha) + l_r \, \dot{\gamma} \, cos \, \alpha}{l_f \, cos(\gamma - \alpha) + l_r \, cos \, \alpha} \tag{14}$$

It should be noted that equation (14) is more accurate if the slip angles are known a priori, a task that is quite difficult as the slip angles are dependent of the vehicle's speed, mass, tire–terrain interaction and articulation angle, in a highly non–linear way. Moreover, in calculating the angular velocity of the tractor, the primary sources of the errors, are due to the time varying parameters $\gamma, \dot{\gamma}, \beta$ and α, as errors in these parameters propagate directly to the states. The variables γ and $\dot{\gamma}$ can be measured with a great accuracy, while an estimation algorithm or a look up table should be utilized for determining the value of the slip parameters.

2.3 Error Dynamics Modeling

Before proceeding with the design of the control scheme, an appropriate model should be initially derived. In the sequel, the error dynamics modeling procedure will be presented. while it should be highlighted that this is the first time in the relevant scientific literature that such a modeling framework, with the ability to take under consideration the effect of slip angles, is being presented.

Based on the assumption that the vehicle develops a steady–state motion turning, or $\dot{\gamma} = 0$, the rates of the orientation change are being provided by: $r_1 = \frac{v_f}{\dot{\theta}_f}$ and $r_2 = \frac{v_r}{\dot{\theta}_r}$. In Figure 3 it is depicted an overview of how the displacement, heading and curvature errors are being defined, between the actual path of the articulated vehicle and the desired one. In this figure, the distance from the vehicle to the reference path displacement, as the angle between the vehicle and the reference are also being displayed.

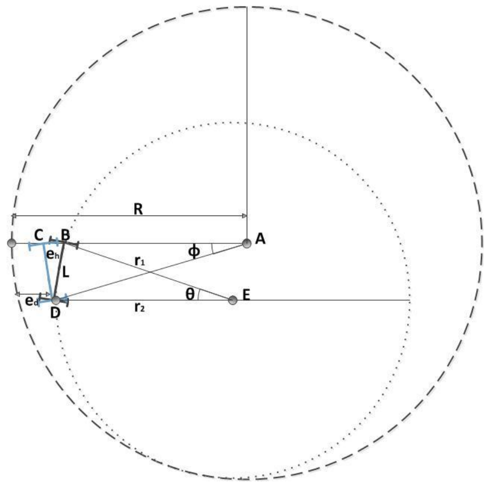

Figure 3: Articulated vehicle following error transformation

In the following derivation, three errors are defined as: a) e_d is the displacement error, b) e_h is the heading error, and c) e_c is the curvature error. Based on [5], the displacement error e_d is the difference between the coordinates of the tractor and the coordinates of the desired circular path. From the triangle (abd), it can be defined that $\theta = \arctan\frac{l}{r}$ and from the triangle (dbc) $e_h = \arctan(\frac{e_d}{l})$. If it is assumed that ϕ and θ are small, l can be defined as $l = r\ \theta$ and $e_d = r\theta e_h$, and by taking the time derivative of this equation yield:

$$\dot{e}_d = v\ e_h \tag{15}$$

The heading error e_h is the orientation difference between the centers of the vehicle and the circular path. A change in the heading error is being defined as $e_h = \theta - \phi$, while the relation between θ and ϕ with respect to l, after small assumptions, is defined as: $(\theta\ r = \phi\ R)$. From this equation it can be derived that $e_h = \frac{\theta\ R - \theta\ r}{R}$, and by taking the time derivative results in: $\dot{e}_h = \dot{\theta} r(\frac{1}{r} - \frac{1}{R})$. The curvature error e_c measures the difference between the vehicle's path circle and the curvature of the trajectory path, and it can be defined as: $e_c = (\frac{1}{r} - \frac{1}{R})$. By utilizing the relations in above equations, the curvature error is being defined as:

$$\dot{e}_h = v\, e_c \tag{16}$$

With the assumption that the velocity and the curvature of the trajectory are constant and by differentiating the curvature equation with respect to time, the rate of the curvature error is being defined as:

$$\begin{aligned}\dot{e}_c \;=\; & \frac{\dot{\gamma}\, l_r cos(\gamma+\beta-\alpha) cos\,\alpha + \dot{\gamma} l_f\, cos(\gamma+\beta-\alpha)\, cos(\gamma-\alpha)}{(l_r\, cos\alpha + l_f\, cos(\gamma-\alpha))^2} + \\ & \frac{\dot{\gamma}\, v\, l_f\, sin(\gamma+\beta-\alpha)\, sin(\gamma-\alpha) - \dot{\gamma}^2\, l_f\, l_r\, cos\,\alpha)\, sin(\gamma-\alpha)}{v(l_r\, cos\alpha + l_f\, cos(\gamma-\alpha))^2} - \\ & \frac{\dot{\gamma}\, l_f\, l_r\, cos(\gamma-\alpha)\, sin(\alpha)\dot{\alpha} + (l_r^2\, cos\alpha\, sin\alpha\, \dot{\alpha})}{v(l_r\, cos\alpha + l_f\, cos(\gamma-\alpha))^2}\end{aligned} \tag{17}$$

Linearizing the error dynamics in equation (17) around the reference path γ yields:

$$\dot{e}_c = \dot{\gamma}\, \frac{(l_f + l_r) + l_f(\gamma^2 + \gamma\beta - 2\gamma\alpha) + l_f(\alpha^2 - \alpha\beta)}{(l_r + l_f)^2} \tag{18}$$

where it can be observed that $\dot{e}_c$ is dependant on the slippage and the rate of change of articulation angles. By performing the following change of variable:

$$\acute{\dot{e}}_c = \dot{e}_c\, - \dot{\gamma}\, \frac{l_f(\gamma^2 + \gamma\beta - 2\gamma\alpha)}{(l_r + l_f)^2} \tag{19}$$

The following state space error dynamics description for the articulated vehicle is being extracted:

$$\begin{bmatrix} \dot{e}_d \\ \dot{e}_h \\ \acute{\dot{e}}_c \end{bmatrix} = \begin{bmatrix} 0 & v & 0 \\ 0 & 0 & v \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_d \\ e_h \\ \acute{e}_c \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{l_f(1+\beta-2\alpha)}{(l_f+l_r)^2} \\ \frac{l_r+l_f(1+\alpha^2-\alpha\,\beta)}{(l_f+l_r)^2} \end{bmatrix} \dot{\gamma} \tag{20}$$

In the case that the articulated vehicle is being driven over terrains that are characterized by different slip angles, multiple larger operating sets of slip angles, with the same characteristics, $a_i \in \mathscr{L}_1$ and $b_i \in \mathscr{L}_2$, with $\mathscr{L}_1$, $\mathscr{L}_2 \in \Re^2$ and $i \in \mathbf{Z}^+$, can be defined, leading to a switching state space description for the model in (20) as it follows:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_i \dot{\gamma} \tag{21}$$

where:

$$\mathbf{x} = \begin{bmatrix} e_d \\ e_h \\ \acute{e}_c \end{bmatrix},\; \mathbf{A} = \begin{bmatrix} 0 & v & 0 \\ 0 & 0 & v \\ 0 & 0 & 0 \end{bmatrix},\; \mathbf{B}_i = \begin{bmatrix} 0 \\ \frac{l_f(1+\beta_i-2\alpha_i)}{(l_f+l_r)^2} \\ \frac{l_r+l_f(1+\alpha_i^2-\alpha_i\,\beta_i)}{(l_f+l_r)^2} \end{bmatrix}$$

and $\mathbf{x} \in \mathscr{X} \subseteq \Re^3$ is the state vector, $\dot{\gamma} \in \mathscr{U} \in \Re$ is the control action, $\mathbf{A} \in \Re^{6\times 6}$, $\mathbf{B_i} \in \Re^{3\times 1}$, and full state feedback is being considered, or $\mathbf{C} = \mathbf{I}_{3\times 3}$.

3 Switching Model Predictive Control Design

MPC is a highly effective control scheme that is able to take under consideration multiplicative system model descriptions, uncertainties, nonlinearities and physical and mechanical constraints in the system model parameters or in the control signals [18]. In the current research effort, the MPC schemes is able to predict future values of the vehicles error dynamics based on the present available information and the current constraints [19]. The MPC control action, which is the rate of articulation angle, is based on a finite horizon continue time minimization of predicted tracking error with constraints on the control inputs and the state variables.

The overall block diagram of the proposed closed loop system is depicted in Figure 4. For efficiently controlling the articulated vehicle, the controller utilizes the current state of motion of the vehicle as well as the next target points of the reference trajectory. The trajectory planner is generating the desired path, while in the sequel this path (planar coordinates) are being translated to displacement, heading and curvature coordinates that act as the reference input for the MPC controller. In the presented methodology for the design of the MP–controller scheme, the mode selector signal of the MPC is the estimated slip angle for the front part of the vehicle using estimation approaches (like extended Kalman filter). The formulation of the MPC is based on: a) the current full state feedback, b) the active constrains on the system, c) the estimated or measured slip angles, and d) the mode selector signal, applies the necessary switching optimal control action.

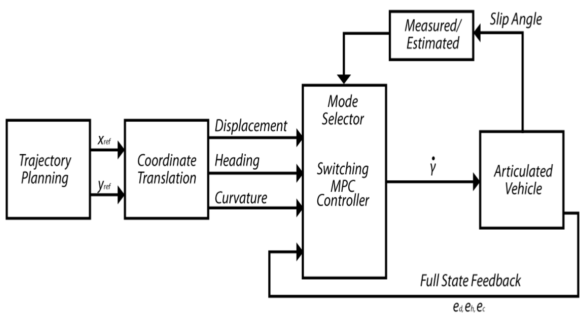

Figure 4: Switching MPC scheme block diagram

The construction of the MP–controller is based on the system description defined in equation (21). The mode selector signal $i \in \mathscr{S}$ with $\mathscr{S} \triangleq \{1,2,\cdots,s\}$ is a finite set of indexes and s denotes the number of switching sub–systems in (22). For polytopic description, Σ is the polytope $\Sigma : Co\{[\mathbf{A}\,\mathbf{B_1}],\cdots,[\mathbf{A}\,\mathbf{B_s}]\}$, Co denotes the convex hull and $[\mathbf{A},\mathbf{B_i}]$ are the vertices of the convex hull. Any $[\mathbf{A},\ \mathbf{B_i}]$ within the convex set Σ is a linear combination of the vertices $\sum_{j=1}^{s}\mu_j[\mathbf{A}\,\mathbf{B_i}]$with $\sum_{j=1}^{s}\mu_j = 1,\ 0 \leq \mu_j \leq 1$. In the presented methodology for the design of the MPC scheme, the mode selector signal is the estimated slip angle. For defining the switching instances, the sets $\mathscr{L}_1$ and $\mathscr{L}_2$ have been discretized into equal operating subspaces. The discretization of the the slip angles operating set, can be formulated by defining multiple nominal values α_0, β_0 and allowing them to take values into neighboring regions of lengths ξ_i and ψ_i. This can be formulated as:

$$\begin{aligned}\mathscr{L}_{1,i} &= \alpha_{0,i}^{\min} = \alpha_{0,i} - \xi_i \leq \alpha_{0,i} \leq \alpha_{0,i}^{\max} + \xi_i = \alpha_{0,i}^{\max} \\ \mathscr{L}_{2,i} &= \beta_{0,i}^{\min} = \beta_{0,i} - \psi_i \leq \beta_{0,i} \leq \beta_{0,i}^{\max} + \psi_i = \beta_{0,i}^{\max}\end{aligned}$$

and by assuming that both slip angles are taking the same values at the same time instances, due to the fact that the length of the articulated vehicle is being considered small, results in ξ_i and ψ_i and for the switching regions:

$$\mathscr{L}_1 = \mathscr{L}_2 = \bigcup \mathscr{L}_{1,i} = \bigcup \mathscr{L}_{2,i}$$

The sets $\mathscr{X}$ and $\mathscr{U}$ specify state and input constraints. Let the set $\mathscr{X}$ contain the $\mathbf{x}$ states that satisfy the following bounding inequality:

$$\mathbf{x}^{\min} = \mathbf{x} - \Delta_1 \leq \mathbf{x} \leq \mathbf{x} + \Delta_1 = \mathbf{x}^{\max}$$

where $\Delta_1 \in \Re_+^{(3,1)}$ is the vector containing the selecting state boundary conditions. The control input bounding set $\mathscr{U}$ can be derived by taking under consideration the mechanical and the physical constraints of the articulated vehicle, as also the preference on aggressive or not maneuvers. These constraints can be also formulated as presented in (23).

$$u^{\min} = u - \Delta_2 \leq u \leq u + \Delta_2 = u^{\max}$$

where $\Delta_2 \in \Re_+$ is the vector containing the selecting control boundary conditions. Let the matrix $\mathbf{H_i}$ be a zeroed 2×2 matrix with its i–th column equal to $[1, -1]^T$, and the other $\mathbf{0}_{2,1}$ i.e. for $i = 2$:

$$\mathbf{H_i} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

Then the previous bounds can be cast in a more compact form as:

$$\begin{bmatrix} \mathbf{H_1} \\ \mathbf{H_2} \end{bmatrix}_{4\times 2} \cdot \begin{bmatrix} \mathbf{x} \\ -- \\ u \end{bmatrix}_{2\times 1} \leq \begin{bmatrix} \mathbf{x}^{\max} + \Delta_1 \\ ------ \\ u^{\max} + \Delta_2 \end{bmatrix}_{4\times 1}$$

These constraints are embedded in the Model Predictive Control computation algorithm in order to compute an optimal controller that counts for the physical and mechanical constraints that restrict the articulated vehicle's motion.

The basic idea of MPC is to calculate a sequence of future control actions in such a way that it minimizes a cost function defined over a predefined prediction horizon. The index to be optimized is the expectation of a quadratic function measuring the distance between the predicted system's output and some predicted reference sequence over the horizon in addition to a quadratic function measuring the control effort. More specifically, the (i)–th MP–controller's objective is to optimize the quadratic cost in (24), while the (i)–th linearized system is within Σ.

Special care must be provided in order to correctly tune the prediction N_p and the control N_c horizon. A long prediction horizon increases the predictive ability of the MPC controller but on the contrary it decreases the performance and demands more computations. The control horizon must also be fine–tuned since a short control horizon leads to a controller that tries to reach the set–point with a few conservative motions, a method that might lead to significant overshoots. On the contrary, a long control horizon produces more aggressive changes in the control action that tend to lead to oscillations. Obviously the tuning of prediction and control horizon is a coupled process. For the aforementioned reasons the control horizon must be chosen short enough compared to the prediction horizon. Additionally, the response of the system can be also shaped using weight matrices on the system outputs, the control action and the control rates.

Each model predictive controller V^i_{MPC} corresponds to the i-th error dynamic model of the articulated vehicle, obtained by solving the following optimization problem $\min J(k)$ with respect to the control moves variations Δu and to the error coordinates, k the discrete time sample index and $J(k)$ defined as:

$$\begin{aligned} J(k) \quad = \quad & \sum_{n=N_w}^{N_p} [\hat{\mathbf{y}}(k+n|k) - \mathbf{r}(k+n|k)]^T \mathbf{Q} [\hat{\mathbf{y}}(k+n|k) - \mathbf{r}(k+n|k)] + \\ & + \sum_{n=0}^{N_c-1} [\Delta u^T(k+n|k) \mathbf{R} \Delta u(k+n|k)] \\ & + \sum_{n=N_w}^{N_p} [u(k+n|k) - \mathbf{s}(k+n|k)]^T \cdot \mathbf{N} [u(k+n|k) - \mathbf{s}(k+n|k)] \end{aligned} \tag{22}$$

where, n is the index along the prediction horizon, N_w is the beginning of the prediction horizon, $\mathbf{Q}$ is the output error weight matrix, $\mathbf{R}$ is the rate of change in control weight matrix, $\mathbf{N}$ is the control action error weight matrix, $\hat{\mathbf{y}}(k+n|k)$ is the predicted system's output at time $k+n$, given all measurements up to including those at time k, $\mathbf{r}(k+n|k)$ is the output set–point profile at time $k+n$, given all measurements up to including those at time k, $\Delta u(k+n|k)$ is the predicted rate of change in control action at time $k+n$, given all measurements up to including those at time k, $u(k+n|k)$ is the predicted optimal control action at time $k+n$, given all measurements up to and including those at time k, and $\mathbf{s}(k+n|k)$ is the input set–point profile at time $k+n$, given all measurements up to and including those at time k.

Once all $V^i_{MPC}(k)$ controllers are computed, the total switching MP–controller is constructed by implementing a switching among the difference controllers in relation with the estimated/measured values of slip angles α and β. The objective of the $i \in \mathscr{S}$ controller is to stabilize the i–th system, while if s increases, then the approximation of the error dynamic modeling is more accurate for a larger part of slip angles and thus allowing the development of more efficient flight control algorithms.

4 Simulation Results

For simulating the efficacy of the proposed control scheme for the problem of path following for an articulated vehicle, over a terrain with varying slip angles, the following vehicle's characteristics have been considered: $l_f = 0.6m$, $l_r = 0.8m$ and constant speed $u = 2m/sec$, values that have been extracted from the real articulated vehicle in Figure 1. Three operating sets for the slip angles have been defined as: $0 \leq |\beta_1| \leq 0.02$, $0.02 < |\beta_2| \leq 0.04$, and $0.04 < |\beta_3| \leq 0.09$, while the bounds on the articulated angle have been defined as $-0.785 \leq \gamma \leq 0.785$, and the bounds on the dynamic errors were $-0.2 \leq \mathbf{x} \leq 0.2$. The proposed controller has been evaluated in a closed circle reference path. The corresponding predictive horizon has been set to $N_p = 10$ and the control horizon as $N_c = 5$ with control interval $0.2sec$.

In Figure 5 the results from path tracking by utilizing switching and non-switching MP–controller under varying slip angles are being presented. As it can been observed there is a significant tracking error, due to the effect of slippage angles deteriorating the overall controller performance.

In Figure 6 the error dynamics and in Figure 7 the articulation angle and articulation angle's rate for the two cases are being presented. It should be noted that: a) in the presented simulation results, harsh switching and big slippage angles have been considered, that simulate the translation of the vehicle under heavy road conditions, and b) the same switching angles have been utilized as the testing scenario in the depicted simulation results.

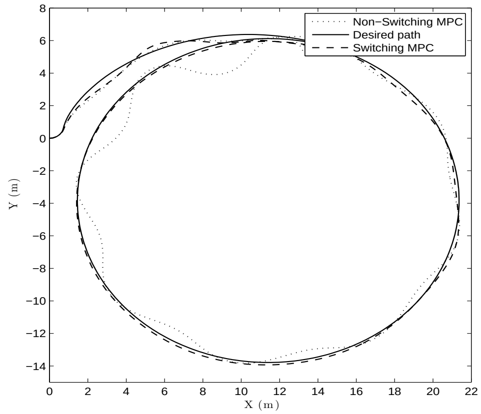

Figure 5: Circle path tracking based on switching MP-controller and non-switching MP–controller under varying slip angles

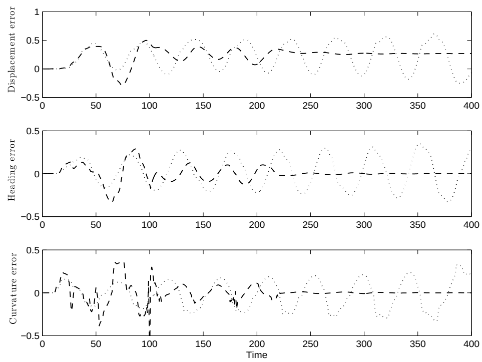

Figure 6: Time evolution of the system error dynamics for both case studies

5 Conclusions

In this article a switching model predictive control scheme for an articulated vehicle under varying slip angles has been presented. The non-linear kinematic model that is able to take under consideration

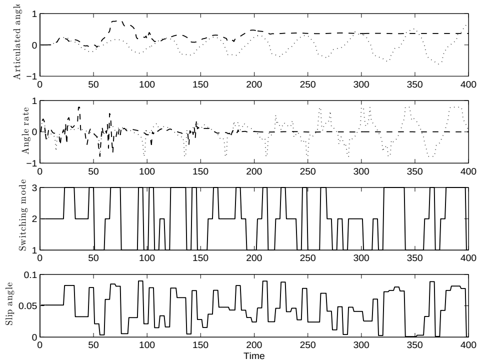

Figure 7: The articulated angle as a control signal, switching mode selector and the slip angle

the effect of the slip angles was extracted and been transformed into an error dynamics model, which in the sequence has been linearized around multiple nominal slip angle cases. The varying slip angle has been considered as the switching rule and a corresponding switching mode predictive control scheme was designed. The simulation results have been presented that prove the efficacy of the overall suggested scheme. Future work includes the application of the proposed scheme in experimental studies.

References

[1] J. Apostolopoulos and L. Gordillo, "Numerical comparison in steering geometry for robotic vehicle by modeling positioning error," *Automation Robot*, vol. 23, pp. 147–159, 2007.

[2] J. Markdahl, G. Bortolin, and U. Andersson, "Traction control for articulated off-road vehicle," in *Master Thesis*, KTH Royal Institute of Technology, 2010.

[3] C. Altafini, "Why to use an articulated vehicle in underground mining operations?" in *International Conference on Robotics and Automation*, Detroit, Michigan, May 1999.

[4] B. Pacejka, "Tire and vehicle dynamics (2nd ed.)," *Society of Automotive Engineers*, vol. 23, pp. 147–159, 2005.

[5] P. Ridley and P. Corke, "Autonomous control of an underground mining vehicle," in *proc. Australian Conference on Robotics and Automation*, Sydney, Nov. 2001.

[6] P. Petrov and P. Bigras, "A practical approach to feedback path control for an articulated mining vehicle," in *Pros. of the 2001 IEEE/RSJ, Int. Conf. on Intelligent Robotics and Systems*, aui, Hawaii, USA, Oct. 2001.

[7] U. Larsson, G. Zell, K. Hyyppa, and A. Wernesson, "Navigating an articulated vehicle and reversing with a trailer," in *IEEE*, 1994.

[8] P. I. Corke and P. Ridley, "Steering kinematics for a center articulated mobile robot," *IEEE Trans. On Robotics and Automation*, vol. 17, no. 2, pp. 215–218, 2001.

[9] D. Piyabongkarn, R. Rajamani, J. Grogg, and J. Lew, "Development and experimental evaluation of a slip angle estimator for vehicle stability control," *IEEE Trans on Control System Technology*, vol. 17, no. 1, pp. 78–88, January 2009.

[10] P. Bigras, P. Petrov, and T. Wong, "A LMI approach to feedback path control for an articulated mining vehicle," in *Electronics Research*, 2002.

[11] P. Ridley and P. Corke, "Load haul dump vehicle kinematic and control," in *Transaction of the ASME*, 2003, pp. 54–59.

[12] A. Hemami and V. Polotski, "Path tracking control problem formulation of an LHD loader," *International Journal of Robotics Research*, vol. 18, no. 5, 1999.

[13] C. Altafini, "A path tracking criterion for an lhd articulated vehicle," in *International Journal of Robotics Research*, May 1999.

[14] R. M. Densantis, "Modeling and path tracking for a load haul dump mining vehiclel," *Transaction of the ASME*, vol. 119, 1997.

[15] P. Bolzern and A. Locatelli, "A comparative study of different solutions to the path-tracking problem for an articulated vehicle," in *Conference on Control Applications*, Glasgow, Scotland, U.K, 2002, pp. 18–20.

[16] S. Scheding, G. issanayake, E. Nebot, and H. Durrant, "Slip Modeling and Aided Inertial Navigation of an LHD," in *International Conference on Robotics and Automation*, New Mexico, 1997.

[17] J. Markdahl, "Traction control for off-road articulated vehicle," Ph.D. dissertation, KTH, Jan 2010.

[18] F. Nunez, S. Navarro, A. Aguado, and A. Cipriano, "State estimation based model predictive control for LHD vehicle," in *17th World Congress, the International Federation of Automatic Control*, Seoul, Korea, 2008.

[19] F. Kuhne, J. Manoel, and W. F. Lages, "Mobile robot trajectory tracking using model predictive control," in *IEEE LARS*, 2005.

PAPER C

Path following for an articulated vehicle based on switching model predictive control under varying speeds and slip angles

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Path following for an articulated vehicle based on switching model predictive control under varying speeds and slip angles

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Abstract: This article is focusing on the problem of path following for an articulated vehicle under varying velocities and slip conditions. The proposed control architecture consists of a switching control scheme based on multiple model predictive controllers, fine tuned for dealing with different operating speeds and slip angles. In the presented analysis for the non–holonomic articulated vehicle, the corresponding kinematic model is being transformed into an error dynamics model, which is linearized around multiple nominal slip angle cases and various operating speeds. The existence of the slipping and varying speed has a significant effect on the vehicle's path following capability and can significantly deteriorate the performance of the overall control scheme. Based on the derived multiple dynamics modeling, the current slip and vehicle's speed are being considered as the signal selector for the proposed switching model predictive control scheme. The efficacy of the proposed controller is being evaluated by an extended set of simulation results.

1 Introduction

In general the articulated vehicles consist of two parts, a front and a rear, linked with a rigid free joint. Each body has a single axle and the wheels are all non–steerable, while the steering action is being performed on the joint, by changing the corresponding articulated angle, between the front and the rear of the vehicle.

In the relative literature, there have been several research approaches for the problem of modeling articulated vehicles, based on the theory of multiple body dynamics [1–3]. Most of these methods contain simple models that: a) are not taking under consideration the effect of the slip angles, as the complexity of the overall problem is being increased and it is more difficult to proceed to the next stage of the control scheme design based on highly non–linear articulated vehicle's dynamics, and b) in these modeling approaches, speed has being considered as a constant, an assumption that it is not globally valid, while at the same time the speed level is also effecting the slippage that the vehicle is being experiencing.

As slip angle is one of the most important factor that degrades the overall performance of the articulated vehicle, various efforts have been proposed until now for estimating or measuring these effects [4, 5], while still the existence of full kinematic models that will integrate the slip angle effect, for the case of articulated vehicles, is obsolete. From a control point of view, there have been proposed many traditional techniques for non–holonomic vehicles, based on error dynamics models without the presence of slip angles. More analytically, in [2] linear control feedback has been applied, while in [4] a Lyapunov based approach has been presented. In [6] a control scheme based on linear matrix inequalities has been presented and in [7] a pole placement technique has been applied. Moreover, in [8] the authors have presented a path tracking controller based on error dynamics, while in [9] the general problem of designing a path following controller for a *n*–rear vehicle, based on non–linear adaptive control has been derived. Finally, classical fundamental problems of motion control for articulated vehicles have been presented in [3, 10] and [11].

The novelty of the proposed article stems from: a) the derivation of an error dynamics modeling framework for the case of an articulated vehicle operating under varying speeds and slip angles, and b) adopting a switching model control scheme that is able to take under consideration the effects of real life constraints on the control input (articulated angle) and mechanical restrictions, while at the same time contains multiple switching model predictive controllers that are being fine tuned to optimize path tracking under varying slip angles and speeds. The overall resulting control scheme is ensuring smooth transition of the control effort as the articulated vehicle is driven with different speeds, over regions of different slippage, while retaining the stability of the articulated vehicle.

2 Articulated Vehicle Modeling

2.1 Kinematic Model Without Slip Angles

First the case of an articulated vehicle, under no effect of slip angles ($\alpha = \beta = 0$) is being presented, as it is depicted in Figure 1, where (x_f, y_f) and (x_r, y_r) denote the coordinates of the front and the rear, p_f and p_r are the corresponding centers of gravity, l_f and l_r are the lengths of the front and rear, while the angles θ_f and θ_r denote the vehicle's part orientation. The (x, y) axes represent the fixed coordinating system, defined as [2]. By examining the vehicle's depicted geometry, as also the relation between the coordinates of p_f and p_r, it can be extracted that:

$$\dot{x}_f = v_f \cos \theta_f \quad (1)$$

$$\dot{y}_f = v_f \sin \theta_f \quad (2)$$

The velocities at the front and the rear parts has the same changing with respect to the velocity at the rigid free joint of the vehicle for the two parts, then can being defined the relative velocity vector equations as:

$$v_f = v_r \cos \gamma + \dot{\theta}_r l_r \sin\gamma \quad (3)$$

$$v_r \sin\gamma = \dot{\theta}_f l_f + \dot{\theta}_r l_r \cos \gamma \quad (4)$$

Where v_f and v_r are the velocities of the front and rear parts respectively, the articulated angle γ is being defined as the difference between the orientation angles θ_f of the front and θ_r of the rear part of the vehicle, by combining the above equations using the relation $\theta_r = \theta_f - \gamma$, the angular velocity $\dot{\theta}_f$ of the front is being calculated as:

$$\dot{\theta}_f = \frac{v_f \sin\gamma + l_r \dot{\gamma}}{l_f \cos\gamma + l_r} \quad (5)$$

The corresponding angular velocities for the front and rear parts, which are being defined as $\dot{\theta}_f$ and $\dot{\theta}_r$ respectively, have different values when ($l_f \neq l_r$) or the vehicle is not driving straight ($\gamma \neq 0$). Finally, the articulated angle γ is being defined as the difference between the orientation angle θ_f of the front and the orientation angle θ_r of the rear part of the vehicle ($\dot{\theta}_r = \dot{\theta}_f - \dot{\gamma}$), or:

$$\dot{\theta}_r = \frac{v_f \sin\gamma - l_f \dot{\gamma} \cos\gamma}{l_f \cos\gamma + l_r} \quad (6)$$

2.2 Kinematic Model Under Slip Angles

In the case of slip angles, the kinematic model of the articulated vehicle can be also formulated from general geometry that includes ideal and slip behavior for the steering angles. The definition of this model has been initially based on the derivation in [5], which included the perturbed factors in the vehicle position as two slip variables β and α, being defined as the front's and rear's slip angles respectively. Under the influence of slip angles, the vehicle's configuration is depicted in Figure 1. where $(r_1,\ r_2)$ are the radiuses from instantaneous centers of velocity for the front and

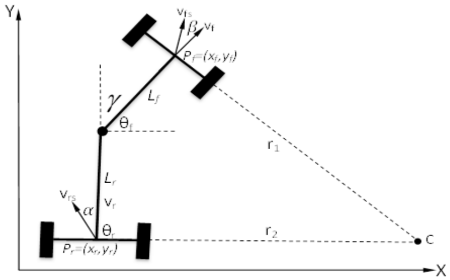

Figure 1: Articulated vehicle modeling configuration under the influence of slip angles

the rear respectively. For the initial center curvature of the trajectory it is being assumed that the vehicle is moving forward without slip conditions. In the following derivation, the unit subscript 's' denotes variables in the slip angle case as they have been defined previously in the non–slip examined case. The kinematic equations, for the motion under the effect of slip angles, for the front part can be formulated as:

$$\dot{x}_{fs} = v_f\ cos\ (\theta_f + \beta) \tag{7}$$
$$\dot{y}_{fs} = v_f\ sin\ (\theta_f + \beta) \tag{8}$$

In this case, the vehicle's motion depends not only on vehicle's velocities, the articulated angle and the vehicle's lengths, but also on the slip angles, while the resulting vehicle's heading is provided by a combination of the slip angles with the orientation angles. The rate of the orientation $\dot{\theta}_{rs}$ can be defined as function of the steering angle γ and both slip angles. Based on the assumption that the vehicle develops a steady–state motion turning, this rate can be provided by the utilization of a virtual center of rotation, depending on the velocity [12]. In the examined case, the front's and rear's speeds can be computed as:

$$v_f\ cos\alpha = v_r\ cos(\gamma - \alpha) + \dot{\theta}_r l_r\ sin(\gamma + \beta - \alpha) \tag{9}$$
$$v_r\ sin(\gamma + \beta - \alpha) = \dot{\theta}_f l_f\ cos\alpha + \dot{\theta}_r l_r\ cos(\gamma - \alpha) \tag{10}$$

while, by solving the above equations, the angular velocities for the front and the rear can be defined as:

$$\dot{\theta}_{fs} = \frac{v_f\ sin(\gamma + \beta - \alpha) + l_r\ \dot{\gamma}\ cos\alpha}{l_f\ cos(\gamma - \alpha) + l_r\ cos\alpha} \tag{11}$$
$$\dot{\theta}_{rs} = \frac{v_f\ sin(\gamma + \beta - \alpha) - l_f\ \dot{\gamma}\ cos(\gamma - \alpha)}{l_f\ cos(\gamma - \alpha) + l_r\ cos\alpha} \tag{12}$$

It should be noted that equations (11,12) are more accurate than equations (7,8), if the slip angles are known a priori, a task that is quite difficult as the slip angles are dependent of the vehicle's velocity, mass, tire–terrain interaction and articulation angle, in a highly non–linear way.

3 Error Dynamics Modeling

Based on the assumption that the vehicle develops a steady–state motion turning, the rates of the orientation change are being provided by: $r_1 = \frac{v_f}{\dot{\theta}_f}$ and $r_2 = \frac{v_r}{\dot{\theta}_r}$. In Figure 2 it is depicted an overview of how the displacement, heading and curvature errors are being defined, between the actual path of the articulated vehicle and the desired one, under the effects of slipping. The distance from the vehicle to the reference path displacement, the angle between the vehicle and the reference are also being displayed.

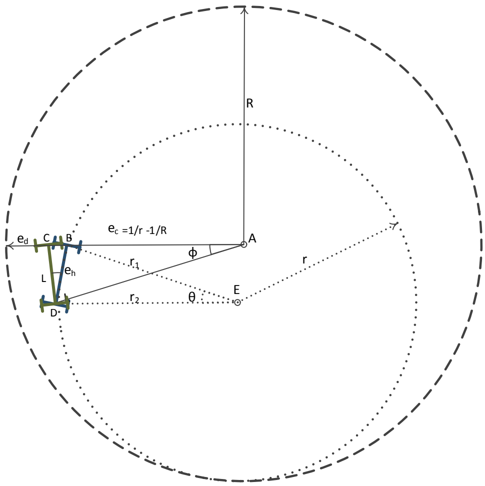

Figure 2: Articulated vehicle following error transformation

Based on [1], in the following derivation, three errors under the effects of slip angles are being defined: a) e_d is the displacement error, b) e_h is the heading error, and c) e_c is the curvature error. The displacement error e_d is the difference between the coordinates of the front and the coordinates of the desired circular path. From the triangle (ABD), if it is assumed that ϕ and θ are small, and can be defined that: $\theta \triangleq (\frac{l_f + l_r\, cos(\gamma+\beta)}{r})$ and from the triangle (DBC), $e_h \triangleq \frac{e_d}{l_f}$, while the error in the displacement is: $e_d = e_h\, r\theta - l_f\, cos(\gamma+\beta)$, and by taking the time derivative of this equation it is derived that:

$$\dot{e}_d = v\, e_h + \dot{\gamma}\, l_f\, sin(\gamma+\beta) \tag{13}$$

The heading error e_h is the orientation difference between the centers of the vehicle and the circular path. A change in the heading error is being defined as: $e_h \triangleq (\theta - \phi) + (\gamma\, sin(\gamma+\beta-\alpha))$, and it is being assumed that: $\theta r \simeq \phi R$. From this equation it can be derived that $e_h = \theta(1-\frac{r}{R}) + (\gamma\, sin(\gamma+$

$\beta - \alpha)$. The curvature error e_c measures the difference between the vehicle's paths and the curvature of the trajectory path, and it can be defined as: $e_c = \frac{1}{r} - \frac{1}{R}$. By utilizing these relations, the curvature error is being defined as:

$$\dot{e}_h = v\, e_c + \dot{\gamma}\, sin(\gamma + \beta - \alpha) + \dot{\gamma}\gamma\, cos(\gamma + \beta - \alpha) \tag{14}$$

With the assumption that the slip angles, the velocity and the curvature of the trajectory are piecewise constant and by differentiating the equation: $e_c = \frac{v\, sin(\gamma+\beta-\alpha)+l_r\, \dot{\gamma}\, cos\alpha}{v\, (l_f\, cos(\gamma-\alpha)+l_r\, cos\alpha)}$ with respect to time, the rate of the curvature error is being defined as:

$$\begin{aligned} \dot{e}_c = \quad & \dot{\gamma} \frac{l_f\, cos(\beta) + l_r\, cos(\alpha)\, cos(\gamma + \beta - \alpha)}{(l_f\, cos(\gamma - \alpha) + l_r\, cos(\alpha))^2} + \\ & \ddot{\gamma} \frac{l_r\, (l_f\, cos(\gamma - \alpha)\, cos(\alpha) + l_r\, cos(\alpha)^2)}{v(l_f\, cos(\gamma - \alpha) + l_r\, cos(\alpha))^2} + \\ & \dot{\gamma}^2 \frac{l_f\, l_r\, sin(\gamma - \alpha)\, cos(\alpha)}{v(l_f\, cos(\gamma - \alpha) + l_r\, cos(\alpha))^2} \end{aligned} \tag{15}$$

Linearizing the error dynamics in the equations (13), (14) and (15) around the reference path with constant articulated and slip angles under small displacement, yields:

$$\dot{e}_d = v\, e_h + \dot{\gamma}\, l_f \beta \tag{16}$$

$$\dot{e}_h = v\, e_c + \dot{\gamma}\, (\beta - \alpha) \tag{17}$$

$$\dot{e}_c = \dot{\gamma}\, \frac{1}{(l_f + l_r)} + \ddot{\gamma}\, \frac{l_r}{v(l_f + l_r)} + \dot{\gamma}^2\, \frac{l_f\, l_r(\gamma - \alpha)}{v(l_f + l_r)^2} \tag{18}$$

where it can be observed that $\dot{e}_c$ is dependant on the slippage and the rate of change of articulation angles. By performing the following change of variable:

$$\acute{\dot{e}}_c = \dot{e}_c\ -\ \dot{\gamma}\, \frac{l_r}{v(l_f + l_r)} \tag{19}$$

The following state space error dynamics description for the articulated vehicle is being extracted:

$$\begin{bmatrix} \dot{e}_d \\ \dot{e}_h \\ \acute{\dot{e}}_c \end{bmatrix} = \begin{bmatrix} 0 & v & 0 \\ 0 & 0 & v \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_d \\ e_h \\ \acute{e}_c \end{bmatrix} + \begin{bmatrix} l_f \beta \\ \frac{l_r + (\beta-\alpha)(l_f+l_r)}{(l_f+l_r)} \\ \frac{1}{(l_f+l_r)} \end{bmatrix} \dot{\gamma} \tag{20}$$

In the case that the articulated vehicle is being driven with varying speed, over terrains that are characterized by different slip angles, multiple piecewise operating sets for the speed and the slip angles, of the same characteristics, can be defined as: $v_i \in \mathscr{L}_1$, $\beta_i \in \mathscr{L}_2$ and $\alpha_i \in \mathscr{L}_3$, with $\mathscr{L}_1$, $\mathscr{L}_2$ $\mathscr{L}_3 \in \Re^3$ and $i \in \mathbf{S}^+$, can be defined, leading to a switching state space description for the model as it follows:

$$\dot{\mathbf{x}} = \mathbf{A}_i \mathbf{x} + \mathbf{B}_i \dot{\gamma} \tag{21}$$

$\mathbf{x} \in \mathscr{X} \subseteq \Re^3$ is the state vector, $\dot{\gamma} \in \mathscr{U} \in \Re$ is the control action, $\mathbf{A_i} \in \Re^{3\times3}$, $\mathbf{B_i} \in \Re^{3\times1}$, and full state feedback is being considered, or $\mathbf{C} = \mathbf{I}_{3\times3}$. The variables γ and $\dot{\gamma}$ can be measured with a great accuracy, while an estimation algorithm or a look up table should be utilized for determining the value of the slip parameters.

4 Model Predictive Control Design

MP–controller has been very successful in practice and its a highly effective control scheme that is able to take under consideration multiplicative system model descriptions, uncertainties, nonlinearities and physical and mechanical constraints in the system model parameters or in the control signals [13]. In the current research effort, the MP–controller schemes is able to predict future values of the vehicles error dynamics based on the present available information and the current constraints [14]. The MP–controller action, which is the rate of articulation angle, is based on a finite horizon continue time minimization of predicted tracking error with constraints on the control inputs and the state variables. The overall block diagram of the proposed closed loop system is depicted in Figure 3. For efficiently controlling the articulated vehicle, the controller utilizes the current state of motion of the vehicle as well as the next target points of the reference trajectory. The path planner is generating the desired path, while in the sequel this path (planar coordinates) are being translated to displacement, heading and curvature coordinates that act as the reference input for the MP–controller.

The formulation of the MP–controller is based on: a) the current full state feedback, b) the active constrains on the system, c) the estimated slip angles, d) the measured velocity and e) the mode selector signal, which is depends on the slipping and the velocity of the vehicle, in order to apply switching optimal control action.

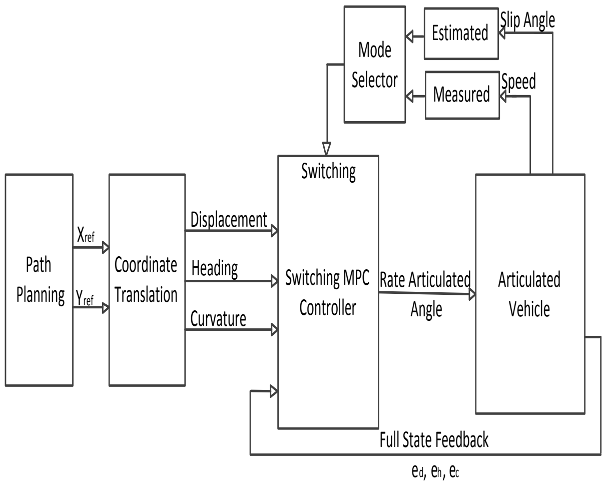

Figure 3: Switching MP–controller scheme block diagram

The construction of the MP–controller is based on the system description defined in equation (21). The mode selector signal $i \in \mathscr{S}$ with $\mathscr{S} \triangleq \{1, 2, \cdots, s\}$ is a finite set of indexes and s denotes the number of switching sub–systems in (24). For polytypic description, Σ is the polytope $\Sigma: \ Co\{[\mathbf{A}_i\, \mathbf{B_i}], \cdots, [\mathbf{A}_i\, \mathbf{B_s}]\}$, Co denotes the convex hull and $[\mathbf{A}_i, \mathbf{B_i}]$ are the vertices of the convex hull. Any $[\mathbf{A_i},\ \mathbf{B_i}]$ within the convex set Σ is a linear combination of the vertices $\sum_{j=1}^{s} \mu_j [\mathbf{A_i}\, \mathbf{B_i}]$with $\sum_{j=1}^{s} \mu_j = 1,\ 0 \leq \mu_j \leq 1$.

In the presented methodology for the design of the MP–controller scheme, only the front slip angle β has been taken under consideration as it has the more significant effect on the behavior of the vehicle than the rear slip angle α, while at the same time α has the opposite sign of β. As a result of this remark the mode selector signals of the MPC are the measured velocity v_i of the vehicle and the estimated slip angle β_i. For defining the switching instances, the sets $\mathscr{L}_1$, and $\mathscr{L}_2$ have been discretized into operating subspaces. The discretization of the speed and slip angle operating sets, can be formulated by defining multiple nominal values $v_0,\ \beta_0$ and allowing them to take values into neighboring regions of lengths ξ_i and ψ_i. This can be mathematically being formulated as:

$$\begin{aligned}\mathscr{L}_{1,\ i} &= v_{0,\ i}^{\min} = v_{0,\ i} - \xi_i \leq v_{0,\ i} \leq v_{0,\ i} + \xi_i = v_{0,\ i}^{\max} \\ \mathscr{L}_{2,\ i} &= \beta_{0,\ i}^{\min} = \beta_{0,\ i} - \psi_i \leq \beta_{0,\ i} \leq \beta_{0,\ i} + \psi_i = \beta_{0,\ i}^{\max}\end{aligned}$$

The sets $\mathscr{X}$ and $\mathscr{U}$ specify state and input constraints. Let the set $\mathscr{X}$ contain the $\mathbf{x}$ states that satisfy the following bounding inequality:

$$\mathbf{x}^{\min} = \mathbf{x} - \Delta_1 \leq \mathbf{x} \leq \mathbf{x} + \Delta_1 = \mathbf{x}^{\max} \tag{22}$$

where $\Delta_1 \in \Re_+^{(3,1)}$ is the vector containing the selecting state boundary conditions. The control input bounding set $\mathscr{U}$ can be derived by taking under consideration the mechanical and the physical constraints of the articulated vehicle, as also the preference on aggressive or not maneuvers. These constraints can be also formulated as presented in (23).

$$u^{\min} = u - \Delta_2 \leq u \leq u + \Delta_2 = u^{\max} \tag{23}$$

where $\Delta_2 \in \Re_+$ is the vector containing the selecting control boundary conditions. Let the matrix $\mathbf{H_i}$ be a zeroed 2×2 matrix with its i–th column equal to $[1,-1]^T$, and the other $\mathbf{0}_{2,1}$ i.e. for $i=2$:

$$\mathbf{H_i} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

Then the previous bounds in (22) and (23) can be cast in a more compact form as:

$$\begin{bmatrix}\mathbf{H_1}\\ \mathbf{H_2}\end{bmatrix}_{4\times 2} \cdot \begin{bmatrix}\mathbf{x}\\ --\\ u\end{bmatrix}_{2\times 1} \leq \begin{bmatrix}\mathbf{x}^{\max} + \Delta_1 \\ ------ \\ u^{\max} + \Delta_2\end{bmatrix}_{4\times 1}$$

These constraints are embedded in the MP–controller computation algorithm in order to compute an optimal controller that counts for the physical and mechanical constraints that restrict the articulated vehicle's motion.

The basic idea of MP–controller is to calculate a sequence of future control actions in such a way that it minimizes a cost function defined over a predefined prediction horizon. More specifically, the (i)–th MP–controller's objective is to minimize the quadratic cost in (24), while the (i)–th linearized system is within Σ. Special care must be provided in order to correctly tune the prediction N_p and control N_c horizons. A long prediction horizon increases the predictive ability of the MP–controller but on the contrary it decreases the performance and demands more computations.

Each model predictive controller V_{MPC}^{i} corresponds to the i-th error dynamic model of the articulated vehicle, obtained by solving the following optimization problem $\min J(k)$ with respect to the control moves variations Δu and the error coordinates, while the discrete time sample index and $J(k)$ defined as:

$$\begin{aligned} J(k) &= \sum_{n=N_w}^{N_p} [\hat{\mathbf{y}}(k+n|k) - \mathbf{r}(k+n|k)]^T \mathbf{Q} [\hat{\mathbf{y}}(k+n|k) - \mathbf{r}(k+n|k)] + \\ &+ \sum_{n=0}^{N_c-1} [\Delta u^T(k+n|k) \mathbf{R} \Delta u(k+n|k)] \\ &+ \sum_{n=N_w}^{N_p} [u(k+n|k) - \mathbf{s}(k+n|k)]^T \cdot \mathbf{N} [u(k+n|k) - \mathbf{s}(k+n|k)] \end{aligned} \tag{24}$$

where, n is the index along the prediction horizon, N_w is the beginning of the prediction horizon, $\mathbf{Q}$ is the output error weight matrix, $\mathbf{R}$ is the rate of change in control weight matrix, $\mathbf{N}$ is the control action error weight matrix, $\hat{\mathbf{y}}(k+n|k)$ is the predicted system's output at time $k+n$, given all measurements up to including those at time k, $\mathbf{r}(k+n|k)$ is the output set–point profile at time $k+n$, given all measurements up to including those at time k, $\Delta u(k+n|k)$ is the predicted rate of change in control action at time $k+n$, given all measurements up to including those at time k, $u(k+n|k)$ is the predicted optimal control action at time $k+n$, given all measurements up to and including those at time k, and $\mathbf{s}(k+n|k)$ is the input set–point profile at time $k+n$, given all measurements up to and including those at time k.

Once all $V^i_{MPC}(k)$ controllers are computed, the objective of the $i \in \mathscr{S}$ controller is to stabilize the i–th system, while if s increases, then the approximation of the error dynamic modeling is more accurate for a larger part of speed and slip angle and thus allowing the development of more efficient control algorithms.

5 Simulation Results

In the presented simulation results it is being assumed that direct measurements for both the velocity and the position of the vehicle can be performed, so no estimation action should be applied to the system, although in a real–life experimental setup, estimation of the slip angles should be performed, by utilizing one estimation algorithm (e.g. Extended Kalman filter).

For simulating the efficacy of the proposed control scheme for the problem of path following for an articulated vehicle, over a terrain with varying speed and slip angles, the following vehicle's characteristics have been considered: $l_f = 0.6m$, $l_r = 0.8m$. Twelve operating sets for the slip angle β have been defined, in combination with three different vehicle's velocities (1, 2 and 3 m/sec), as it has been depicted in Table 1.

Table 1: Active Regions for the Switching MPC

MPC No.	Velocity(m/s)	Front slip (rad)
1	$0 \leq v < 1$	0.00$\leq \beta <$0.02
2	$0 \leq v < 1$	0.02$\leq \beta <$0.04
3	$0 \leq v < 1$	0.04$\leq \beta <$0.06
4	$0 \leq v < 1$	0.06$\leq \beta <$0.08
5	$1 \leq v < 2$	0.00$\leq \beta <$0.03
6	$1 \leq v < 2$	0.03$\leq \beta <$0.05
7	$1 \leq v < 2$	0.05$\leq \beta <$0.07
8	$1 \leq v < 2$	0.07$\leq \beta <$0.09
9	$2 \leq v \leq 3$	0.00$\leq \beta <$0.04
10	$2 \leq v \leq 3$	0.04$\leq \beta <$0.08
11	$2 \leq v \leq 3$	0.08$\leq \beta <$0.12
12	$2 \leq v \leq 3$	0.12$\leq \beta <$0.16

As a result the overall lookup table that rules the switching signal for defining the j active MPC control ($s = 12$) is presented in Figure 4, while the total switching MP–controller is constructed by implementing a switching among the difference controllers in relation with the values of slip angle β and the velocity v respectively.

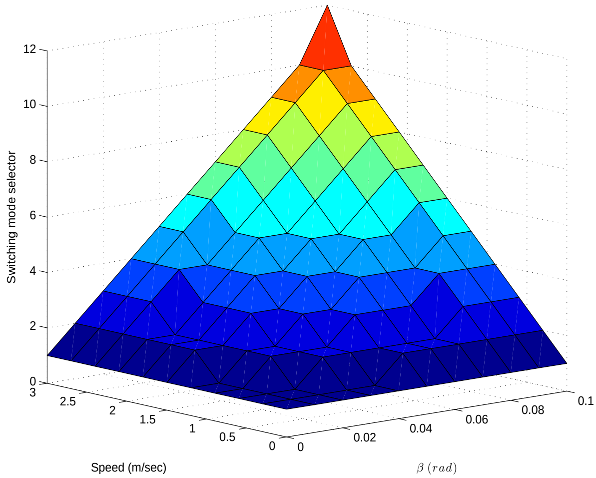

Figure 4: Switching mode selector for MP–controllers

The constraints imposed on the input and output variables, has been defined as it follows: the bounds on the articulated angle have been defined as $-0.785 \leq \gamma \leq 0.785$ (rad), and the bounds on the error dynamics has been set for the displacement error $-0.2 \leq e_d \leq 0.2$. The parameters for the twelve MP–controllers for the state and control action weighings matrices have been fine tuned separately according to the current active state space model. Moreover, in the simulation studies, correlation between the articulated angle γ and the α slip angle, has been considered (derived from the geometry of the vehicle), denoted as: $\alpha = \arcsin(\frac{-l_r}{l_f+l_r}\sin\gamma)$.

The cost function definition provides several crucial pieces of information, which are: length of the prediction horizon, variables included in this cost function and also the weights applied to each of the two components of the cost function. In the presented simulation results, the prediction horizon has been set to $N_p = 10$ and the control horizon to $N_c = 5$, with a control interval $(0.2\ sec)$ for the all controllers.

The effectiveness of the proposed switching MPC scheme will be evaluated on tracking a circular reference path, while the significance and the important deterioration of the vehicle's response, under slip angles, has been already presented in [15]. In the presented simulations, a comparison will be performed between closed loop path tracking results, obtained from the proposed switching MPC scheme and the results obtained by having only one MPC fine tuned controller, under the existence of varying velocities and slip angles, as it is being presented in Figure 5.

From Figure 5 it is clear that the proposed switching MPC has the merit of adapting to the varying type of velocity and articulated angle and achieve a good path tracking performance, while the utilization of only one fixed MP–controller fails to track the circular reference path. In this simulation result, the fixed 2^{nd} MP–controller has been utilized, as this is the controller that had the best simulation output under constant speed $(1\ m/sec)$ and varying slip angles, while the utilization of different fixed type controllers has resulted in a system instability.

In Figure 6 the time evolution of the error corresponding dynamics,$(e_d,\ e_h\ and\ e_c)$, from the

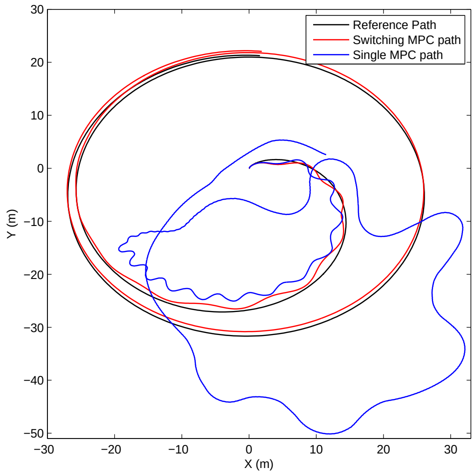

Figure 5: Circle path following based on switching MP–controller and single MP-controller under varying velocities slip angles

path tracking response in Figure 5 is being presented, where it can be also observed that there is a significant error dynamic state tracking, for the case that the fixed MP–controller is being selected.

In Figure 7 the corresponding responses for the simulated time varying velocity, and slip angle, for the path tracking response in Figure 5 are also being presented, with respect to the resulting mode selector signal, from Figure 4. In addition in this figure, the response of the control effort (rate of the articulated angle $\dot{\gamma}$ and the pure response of the articulated angle is being also depicted. From the obtained simulation results, the superiority of the proposed switching MPC scheme is clear as it is able to achieve a good tracking of the reference path, under varying velocities and slip angles, a problem that it is highly linked to the real–life application.

Finally, it should be noted that: a) in the presented simulation results, harsh switching and big slippage angles have been considered, that simulate the translation of the vehicle under heavy road conditions, and b) the same switching angles have been utilized as the testing scenario in all the depicted simulation results (single and switching MP–controller).

6 Conclusions

In this article a switching model predictive control scheme for an articulated vehicle under varying slip angles and different velocities has been presented. For the non–holonomic articulated vehicle, the non–linear kinematic model that is able to take under consideration the effect of the slip angles was extracted and been transformed into an error dynamics model. Based on the derived multiple error dynamic models, the varying slip angle has been considered as the switching rule and a corresponding switching mode predictive control scheme was designed. Simulation results have been presented that prove the efficacy of the overall suggested scheme.

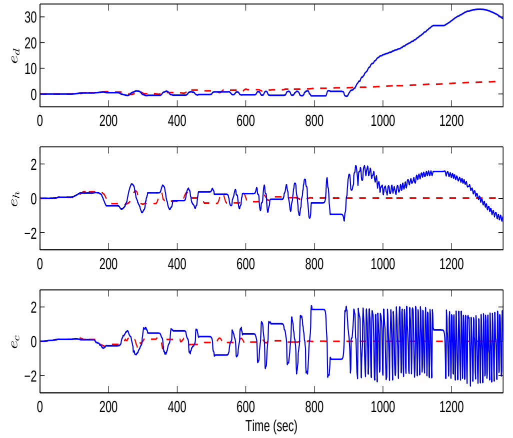

Figure 6: Time evolution of the system's states (error dynamics) with switching MP–controller and single MP–controller

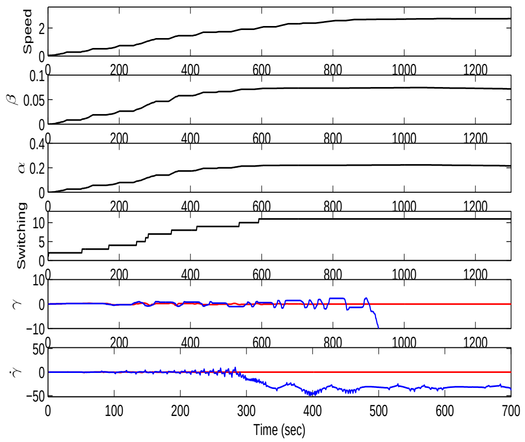

Figure 7: Corresponding varying velocity, slip angle, mode selector signal and control effort for switching and single MP-control

References

[1] P. Ridley and P. Corke, "Autonomous control of an underground mining vehicle," in *proc. Australian Conference on Robotics and Automation*, Sydney, Nov. 2001.

[2] P. Petrov and P. Bigras, "A practical approach to feedback path control for an articulated mining vehicle," in *Pros. of the 2001 IEEE/RSJ, Int. Conf. on Intelligent Robotics and Systems*, aui, Hawaii, USA, Oct. 2001.

[3] U. Larsson, G. Zell, K. Hyyppa, and A. Wernesson, "Navigating an articulated vehicle and reversing with a trailer," in *IEEE*, 1994.

[4] D. Piyabongkarn, R. Rajamani, J. Grogg, and J. Lew, "Development and experimental evaluation of a slip angle estimator for vehicle stability control," *IEEE Trans on Control System Technology*, vol. 17, no. 1, pp. 78–88, January 2009.

[5] S. Scheding, G. issanayake, E. Nebot, and H. Durrant, "Slip Modeling and Aided Inertial Navigation of an LHD," in *International Conference on Robotics and Automation*, New Mexico, 1997.

[6] P. Bigras, P. Petrov, and T. Wong, "A LMI approach to feedback path control for an articulated mining vehicle," in *Electronics Research*, 2002.

[7] P. Ridley and P. Corke, "Load haul dump vehicle kinematic and control," in *Transaction of the ASME*, 2003, pp. 54–59.

[8] A. Hemami and V. Polotski, "Path tracking control problem formulation of an LHD loader," *International Journal of Robotics Research*, vol. 18, no. 5, 1999.

[9] C. Altafini, "A path tracking criterion for an lhd articulated vehicle," in *International Journal of Robotics Research*, May 1999.

[10] R. M. Densantis, "Modeling and path tracking for a load haul dump mining vehiclel," *Transaction of the ASME*, vol. 119, 1997.

[11] P. Bolzern and A. Locatelli, "A comparative study of different solutions to the path-tracking problem for an articulated vehicle," in *Conference on Control Applications*, Glasgow, Scotland, U.K, 2002, pp. 18–20.

[12] J. Markdahl, G. Bortolin, and U. Andersson, "Traction control for articulated off-road vehicle," in *Master Thesis*, KTH Royal Institute of Technology, 2010.

[13] F. Nunez, S. Navarro, A. Aguado, and A. Cipriano, "State estimation based model predictive control for LHD vehicle," in *17th World Congress, the International Federation of Automatic Control*, Seoul, Korea, 2008.

[14] F. Kuhne, J. Manoel, and W. F. Lages, "Mobile robot trajectory tracking using model predictive control," in *IEEE LARS*, 2005.

[15] T. Nayl, G. Nikolakopouls, and T. Gustafsson, “Kinematic modeling and extended simulation studies of a load hull dumping vehicle under the presence of slip angles,” in *IASTED International Conference on Modelling, Simulation and Identification*, Pittsburgh, USA, 2011, pp. 344–349.

PAPER D

On line Dynamic Smooth Path Planning for an Articulated Vehicle

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On line Dynamic Smooth Path Planning for an Articulated Vehicle

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Abstract: This article proposes a novel online dynamic smooth path planning scheme based on a bug like modified path planning algorithm for an articulated vehicle under limited and sensory reconstructed surrounding static environment. In the general case, collision avoidance techniques can be performed by altering the articulated steering angle to drive the front and rear parts of the articulated vehicle away from the obstacles. In the presented approach factors such as the real dynamics of the articulated vehicle, the initial and the goal configuration (displacement and orientation), minimum and total travel distance between the current and the goal points, and the geometry of the operational space are taken under consideration to calculate the update on the future way points for the articulated vehicle. In the sequel the produced path planning is being online and iteratively smoothen by the utilization of Bezier lines before producing the necessary rate of change for the vehicle's articulated angle. The efficiency of the proposed scheme is being evaluated by multiple simulation studies.

1 Introduction

Recently, there have been significant advances in designing automated articulated vehicles mainly for their utilization in the mining industry, where the aim has been the overall increase of the production, while making the working conditions for the human operators safer [1]. In most of the cases, these vehicles are remotely operated, while there is a continuous trend for increasing the autonomy levels, especially in the area of path planning and obstacle avoidance as the vehicles need: a) to perceive the changing environment, based on the onboard sensory systems and b) autonomously plan their route towards the final objective [2].

For the classical task of path planning, with an obstacle detection and avoidance capability, the simplest technique to solve the problem is the altering of the vehicle's orientation, while predicting a non collision path, based on the vehicle's kinematic model, the sensing range and the safety range. In this approach a finite optimal sequence of control inputs, according to the initial vehicle position and the desired goal point is being generated, which is able to take under consideration positioning and measuring uncertainties, such that the collision with any obstacle at a given future time never occurs.

From another point of view, path planning can be divided in two main categories according to the assumptions of: a) global approaches where it is being assumed that the map is a priori available, and b) a partially known and reconstructed surrounding environment based on reactive approaches, which utilizes sensors like infrared, ultrasonic and local cameras. Characteristic examples of the first case are the Road–Map algorithm [3], the Cell Decomposition [4], the Voronoi diagrams [5], the Occupancy Grinds [6] and the new Potential Fields techniques [7], while in most of the cases, a final step of smoothing the produced path curvatures, by the utilization of Bezier curves is being utilized [8].

For the second case of a partially known and online reconstructed environment, the Bug family algorithms are well known mobile vehicle navigation methods for local path planning based on a minimum set of sensors and with a decreased complexity for online implementation [9]. One of the most commonly utilized path planning algorithm in this category is the Bug1 and Bug2 [10]. Bug1 algorithm exhibits two behaviors; motion to goal with boundary following and a corresponding hit point and leave point, while Bug2 algorithm presents similar behaviors like the Bug1 algorithm, except from the fact that it tries to follow the fixed line from a start point to the goal, during obstacle avoidance. Other Bug algorithms that also incorporate range sensors are TangentBug [11], DistBug [12] and VisBug [13]. Tangent Bug algorithm is an improvement of the Bug2 algorithm since it is able to determine the shorter path to the goal using a range sensor with a 360^o infinite orientation resolution. DistBug has a guaranteed convergence and will find a path if one exists, while it requires the perception of its own position, the goal position and the range sensory data [14]. The VisBug algorithm, needs global information to update the value of the minimum distance to the goal point, during the boundary following and for determining the completion of a loop during the convergence to the goal. In all the presented path planning algorithms, the vehicle is being modeled as a point within the world space, without any constraint in the movements, while the actual kinematics of the vehicle, which is important especially in the case of non–holonomic vehicles are being neglected.

The novelty of this article stems from the proposal of a new bug like path planning algorithm based on the dynamic model of an articulated vehicle, which is able to consider: a) the physical constraints of the vehicle, b) proper obstacle detection and avoidance, and c) smooth path generation based on an online Bezier lines processing of the produced way points. In the presented approach the solution to the path planning problem is generated online, based on partial and online sensory information of the vehicle's surrounding environment, while the path is being calculated by solving the inverse kinematic problem of the articulated vehicle or by calculating the optimal articulation angle. Moreover, as in the case of all the exploration and final goal seeking algorithms, it is assumed that the vehicle is constantly aware of the final goal coordinates. During the convergence to this goal and based on the limited range sensing of the surrounding environment, the vehicle is able to detect and avoid obstacles, while continue converging to the optimum goal. This approach is providing an online and sub optimal solution, when compared with the global path planning techniques, and it can be directly applied to the case of articulated vehicles. As it has been applied in the previous path planning algorithms for the case of a priori known space configuration, in the proposed scheme, the Bezier curves are being also utilized for filtering the produced way points and thus guarantee for an online smooth path planning due to the Bezier's line property of continuous higher-order derivatives.

The rest of the article is organized as it follows. In Section 2 the model of the articulated vehicle and the corresponding state space equations will be presented. In Section 3 the proposed novel scheme for smooth path planning and obstacle avoidance based on the articulated vehicle's dynamics will be introduced, while in Section 4 multiple simulation results will be depicted that prove the efficacy of the path planning scheme in different test cases. Finally, the concluding remarks are provided in Section 5.

2 Articulated Vehicle Model

An articulated vehicle is constructed by two parts, a front and a rear, linked with a rigid free joint, while each body has a single axle and the wheels are all non–steerable, with the steering action to be performed on the joint, by changing the corresponding articulated angle γ between the front and the rear of the vehicle [15] as it being also presented in Figure 1.

The main assumptions to derive the kinematic model of the articulated vehicle are: a) the steering angle γ remains constant under small displacement, b) dynamical effects due to low speed, like tire

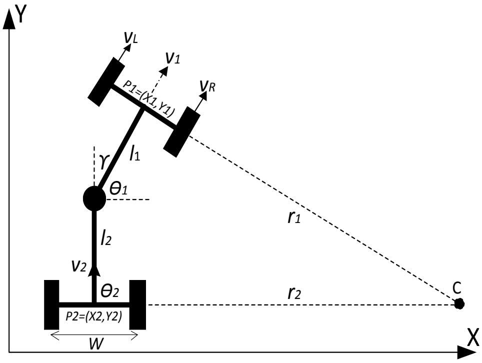

Figure 1: Articulated vehicle's geometry.

characteristic, friction, load and breaking force are being neglected, c) it's assumed that the vehicle moves on a plane without slipping effects, during low-level control, the vehicle's velocities are bounded within the maximum allowed velocities, which prevents the vehicle from slipping, and c) each axle is composed of two wheels and when replaced by a unique wheel, can get:

$$\dot{X}_1 = V_1\, cos\, \theta_1 \tag{1}$$

$$\dot{Y}_1 = V_1\, sin\, \theta_1 \tag{2}$$

The steering angle γ is being defined as the difference between the orientation angles of the front θ_1 and the rear parts θ_2 of the vehicle.

The velocity V_1 at the front and V_2 at the rear parts have the same changing with respect to the velocity at the rigid free joint of the vehicle, and it can be defined by the relative velocity vector equations as it follows:

$$V_1 = V_2\, cos\, \gamma + \dot{\theta}_2 l_2\, sin\gamma \tag{3}$$

$$V_2\, sin\gamma = \dot{\theta}_1 l_1 + \dot{\theta}_2 l_2\, cos\, \gamma \tag{4}$$

where $\dot{\theta}_1$, $\dot{\theta}_2$ and l_1, l_2 are the angular velocities and the lengths of the front and rear parts of the vehicle respectively. By combining these equations it yields:

$$\dot{\theta}_1 = \frac{V_1\, sin\gamma + l_2\, \dot{\gamma}}{l_1\, cos\gamma + l_2} \tag{5}$$

while the angles γ and θ_1 can be measured with a great accuracy. For the case that there is a steering limitation for driving the rear part, according to the coordinates of the point $P_2 = (X_2, Y_2)$, the geometrical relationship between P_1 and P_2 is provided by:

$$X_2 = X_1 - l_1 cos\theta_1 - l_2 cos\theta_2 \tag{6}$$

$$Y_2 = Y_1 - l_1 sin\theta_1 - l_2 sin\theta_2 \tag{7}$$

The realistic dynamic motion behavior of the articulated vehicle with initial parameters $[X_r\ Y_r\ \theta_r\ \gamma_r]$, is depicted in Figure 2, where the vehicle is requested to reach the goal destination with a specific orientation. As it can be observed, when the dynamics of the vehicle are being incorporated the motion and the overall behavior of the vehicle significantly deviates from the case where the vehicle is being considered of having the dynamics of an unconstraint point and this is one of the major contributions of this article.

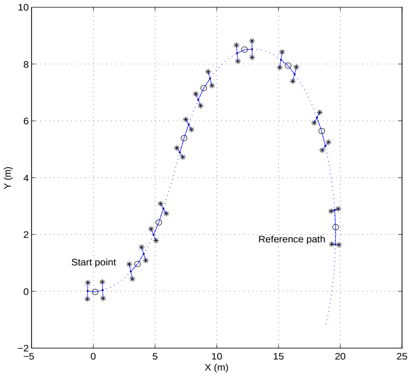

Figure 2: Realistic dynamic motion behavior of the articulated vehicle starting at $[0\ 0\ 0\ -10^0]$ *with* $V = 1m/s$ and $\dot{\gamma} = 0^0$ for the first 5sec of movement, while $\dot{\gamma} = 3.5^0$ for the next 6 *sec* to reach the goal point at $[20\ \ 2]$. The vehicle dimensions are $l_1 = l_2 = 0.6m$ and $W = 0.58m$.

The state parameters of the articulated vehicle are; $\mathbf{X} = [X\ Y\ \theta\ \gamma]^T$ and the manipulated variables are $\mathbf{u} = [V\ \dot{\gamma}]^T$, while the kinematic model of the articulated vehicle, in a state space formulation can be written as it follows:

$$\begin{bmatrix} \dot{X} \\ \dot{Y} \\ \dot{\theta} \\ \dot{\gamma} \end{bmatrix} = \begin{bmatrix} cos\theta & 0 \\ sin\theta & 0 \\ \frac{sin\gamma}{l_1 cos\gamma + l_2} & \frac{l_2}{l_1 cos\gamma + l_2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V \\ \dot{\theta}_{\gamma}, \end{bmatrix} \tag{8}$$

where $\dot{\theta}_{\gamma}$ is the rate of change for the articulated angle.

3 On line Smooth Path Planning for an Articulated Vehicle

The introduced path planning algorithm can be applied for the objective of moving a vehicle from a starting point to the goal point, while detecting and avoiding identified obstacles based on the real vehicles dynamic equations of motion. As a common property of the Bug like algorithms, the proposed scheme initially faces the vehicle towards the goal point, which it is being assumed to be a priori known. In the proposed path planning module it is also being assumed that the vehicle is able to online sense the surrounding environment based on the available sensory systems. The proposed scheme is able to replan the produced path, by generating new way points, after the identification of an obstacle and produce proper path deformations that need to be done for avoiding it. In all these cases the produced set of new way points are utilized as control points for a Bezier curve algorithm for online smoothing of the suggested path. The overall proposed concept of the novel path planning algorithm is being presented in the following Figure 3.

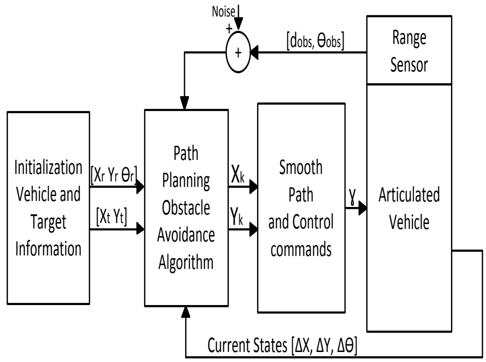

Figure 3: Block diagram of the proposed path planning algorithm based on the nonlinear articulated kinematic model.

As it can be observed from this diagram the algorithm starts by defining the current position and orientation of the vehicle, denoted by $[X_r,\ Y_r,\ \theta_r]$ and the final goal position denoted by $[X_g,\ Y_g,\ \theta_g]$. Based on the onboard sensory system, the vehicle identifies the surrounding environment and obstacles and generates the way points for reaching the goal destination. In the sequel the way points are been smoothen by the utilization of Bezier filtering, while as a last step and based on the vehicles dynamics, an open loop control signal (articulated angle) is being generated to guide the vehicle. In the presented approach it is also assumed that the system is fully observable and good and timely available measurements can be provided for the displacement and orientation of the vehicle.

The assumed sensory system is able to detect the obstacles and the surrounding environment, measure the distance of the articulated robot from the obstacle $d_{obs} \in \Re$, while a sensing radius $\theta_{obs} \in \Re$ is being considered, reflecting real life sensing limitations. In the presented approach all the obstacles and the surrounding environment are being considered as point clouds in a 2-dimensional space, while the overlapping obstacles are being clustered and represented by a single and unified obstacle Figure 4.

The overall flowchart diagram for path planning and obstacle avoidance for the case of an articulated vehicle is being depicted in Figure 5, while is can be summarized as it follows:
[Step 1: Initialization] Define initial $[X_r\ Y_r\ \theta_r\ \dot{\gamma}_r]$ and goal $[X_g\ Y_g]$, define the articulated vehicle's specific parameters V, d_{min}, θ_{min}, d_{obs}, θ_{obs}, the path update rate defined as T and the vehicle's mechanical and physical constraints that needs to be taken under consideration. Set $[X_k\ Y_k\ \theta_k\ ] = [X_r\ Y_r\ \theta_r\ \dot{\gamma}_r]$, with $k \in Z^+$ the sample index.
[Step 2: Path Update] Utilize Equations (1), (2), (6), (7) and (8) to update the coordinates of the next way point as:

$$[X_{k+1}\ Y_{k+1}\ \theta_{k+1}] = [X_k\ Y_k\ \theta_k] + T * [\Delta X\ \Delta Y\ \Delta\theta]$$

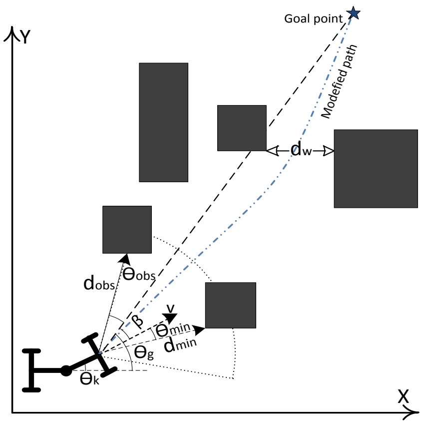

Figure 4: Notations and overall concept of the proposed path planning algorithm.

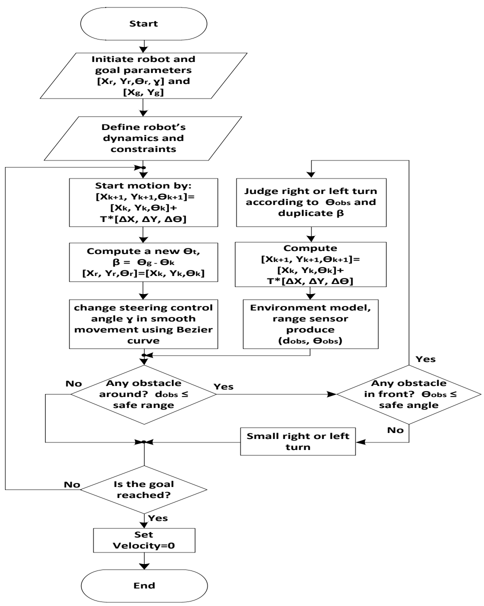

Figure 5: Main flowchart of path planning motion.

calculate:

$$\begin{aligned}\theta_g &= \arctan\frac{Y_{k+1}-Y_g}{X_{k+1}-X_g}\\ \beta &= \theta_g-\theta_k\end{aligned}$$

to produce γ with θ_g the angle between the line that connects the center of gravity of the vehicle's front part to the goal point and the X axis, while β is the difference angle among the vehicle's orientation angle, with the X axis and θ_g. During the application of this step, constraints can be imposed on the articulated vehicle just by bounding the allowable articulation within $\gamma^+ \leq \dot{\gamma} \leq \gamma^-$, with $^+$ and $^-$ representing the maximum and the minimum bounds on the articulated angle.

[Step 3: Obstacle Avoidance] The obstacle avoidance strategy becomes active when the safety conditions d_{min} and θ_{min} are satisfied. This can be evaluated by calculating the update of the distance from the obstacle and the obstacle's angle by:

$$\begin{aligned}Dis_{obs} &= \sqrt{(X_{k+1}-X_{obs})^2+(Y_{k+1}-Y_{obs})^2}\\ \theta_{obs} &= \arctan\frac{Y_{k+1}-Y_{obs}}{X_{k+1}-X_{obs}}\\ \theta_{obs,g} &= \theta_{obs}-\theta_g\end{aligned}$$

while in the case that the following conditions are true:

$$\begin{aligned}&(Dis_{obs}<d_{min}) \quad AND \quad (\theta_{obs,g}<\theta_{min})\\ &\qquad\qquad\qquad OR\\ &(Dis_{obs}<d_{min}) \quad AND \quad (\theta_{obs,g}<-\theta_{min})\end{aligned}$$

the changing in the steering angle is being duplicated and Step 3 is repeated again till the above condition is false and the algorithm continues from Step 2 or the bounds on the articulated angle cannot meet and the algorithm jumps to Step 4.

[Step 4: Reaching Final Goal] If $[X_{k+1}\ Y_{k+1}] = [X_g\ Y_g] \pm Dis_{tolr}$, set $velocity = 0$ and the path planning algorithm has been terminated. Otherwise, the algorithm jumps to Step 2 and the whole process is being repeated in order to avoid collisions with the obstacles until vehicle reaches the goal tolerance distance.

During the execution of the proposed path planning algorithm, and especially Step 2, the proposed path planning algorithm always smoothes the produced way points by the utilization of Bezier curve filtering. The mathematical formulation of the applied Bezier smothering is denoted as:

$$\mathbf{B}(t)=\sum_{i=0}^{n}\binom{n}{i}\,(1-t)^{n-i}\,t^i\,P_i\,,\ \ t\in[0,1] \tag{9}$$

The number n of the considered control points for the Bezier curve generation plays a significant role in the final shape of the produced smooth path as it can be observed from Figure 6, where multiple Bezier lines are being displayed with respect to different number of control points. A n degree Bezier line always passes through the first and last control points and it can be proved that it always lies within the convex hull of the control points, while being tangent to the lines connecting the way points [16].

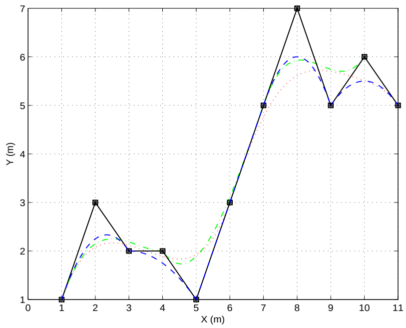

Figure 6: Bezier curves based on different number of control points resulting in different paths, (Black line-solid path produced from the generated way points, red line-dots for 6 points, green line-dash dot for 4 points and the blue line-dash for 3 points).

4 Simulation Results

For simulating the efficacy of the proposed path planner, the following articulated vehicle's characteristics have been considered: $l_1 = l_2 = 0.6m$, $W = 0.58m$, while the vehicle's speed is constant and equal to $1m$sec. Moreover, the constraints imposed on the articulated angle γ have been defined as ± 0.523 *rad* and random measurement Gaussian noise with a fixed variance was added to all measurements of the range sensor to simulate the real life measurement distortion.

The effectiveness of the proposed algorithm will be evaluated in arenas of different types and dimensions. More analytically, the algorithm is simulated on three types of environments with different obstacle configurations, where the vehicle and obstacle geometry is described in a 2D workspace. The obtained outputs of the path planning solutions from the indicated starting points to the goal points is being depicted in Figures 7, 8 and 9, displaying cases with the same sensing radius $d_{obs} = 3m$ and various d_{min}.

As it can be observed in all the examined cases the vehicle is able to avoid all the obstacles, including the bounding surrounding (e.g. walls), which can also be considered as obstacles without loss of generality. In the presented simulations, the articulated vehicle is achieving to reach the reference final goal, independently of the initial vehicle's orientation, while in all the simulations the safety radius has been also displayed with the red circle notation. The basic assumption in all these simulations is that the articulated vehicle, in every time instant, is aware of the coordinates of the final goal and thus the path is being tuned in every step based on the identified obstacles, while the vehicle explores the surrounding environment towards the final goal. In an obstacle free environment, the optimal solution to this problem would have been a straight line connecting the initial with the goal point, a case that can be easily identified in the presented simulation in Figure 7. The online identification of obstacles produces distortions from following the straight line, connecting the robot with the final goal point, while the sensing and safety radius are having a major effect on the path calculation. As it can observed in Figure 10, the safety radius plays a very significant role in shape of the path. In this Figure, different paths with different safe distances d_{min} and with the same sensing radius d_{obs} are being presented. In the case that the vehicle is moving in a bounded space, the selection

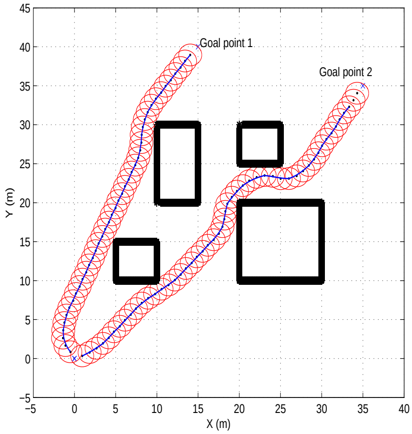

Figure 7: Different shape obstacles placed in the workspace. During these simulations the vehicle starts from different initial angles at $[0,\ 0,\ 120^0,\ 7.5^0]$ and $[0,\ 0,\ 20^0,\ 7.5^0]$ to reach the goal points located at $[15,\ 40]$ and $[35,\ 35]$ respectively, with safe distance $d_{min} = 3m$, $d_{obs} = 3m$.

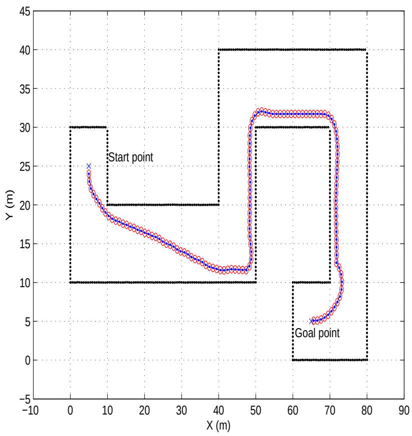

Figure 8: Path planning in an arena having boundaries on both sides of the road a fact that restricts the articulated vehicle motions. During this simulation scenario the vehicle is starting from the initial posture $[5,\ 25,\ -90^o,\ 7.5^o]$ and the goal is located at $[65,\ 5]$ and $d_{min} = 0.5m$, $d_{obs} = 3m$.

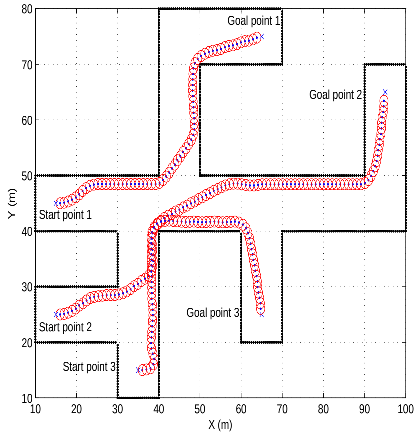

Figure 9: Path planning in an arena having more complicated boundaries on both sides of the road, with noise measurement. The scenario is starting from different initial postures [15, 45], [15, 25] and [35, 15] with initial $[\theta,\ \gamma] = [10^o,\ 7.5^o]$ and the goals are located at [65, 75], [95, 65] and [65, 25] respectively with $d_{min} = 1.0m$, $d_{obs} = 3m$.

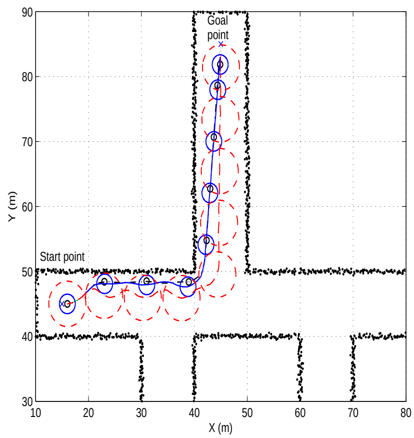

Figure 10: During these simulations the vehicle is starting from the initial posture $[15,\ 45,\ 0,\ 5^0]$ and the goal point is located at [45, 85], while during movement different safe distances d_{min} have been utilized as 0.5, 1.5 and 3.5m, with the same $d_{obs} = 3m$

of a relatively big safety radius introduces oscillations in the translation of the robot due to sequential safety violations that produce corresponding change in the direction of the vehicle for avoiding the obstacle. In case that small safety radius are being selected, this effect is being vanished and smooth and shorter, non oscillatory, paths can be produced.

Finally it should be stated that the arenas in Figures 8 and 9 are typical realistic examples of areas where articulated vehicles operate, as the mine tunnels and the civil roads are. In the presented simulations the consideration of the articulated vehicle's dynamic motion is obvious especially in the time instances where the vehicle is turning towards the goal and while performing at the same time obstacle avoidance. This effect is of paramount importance for the case of articulated vehicles as classical point dynamic approaches in path planning will obviously results in non-realistically achievable paths that would directly lead to collisions.

5 Conclusions

In this article a novel online dynamic smooth path planning scheme based on a bug like modified path planning algorithm for an articulated vehicle under limited and sensory reconstructed surrounding static environment has been proposed. In the presented approach factors such as the real dynamics of the articulated vehicle, the initial and the goal configuration, the minimum and total travel distance between the current and the goal points, the geometry of the operational space, and the path smothering approach based on Bezier lines have been taken under consideration to produce a proper path for an articulated vehicle, which can be followed by correspondingly altering the vehicle's articulated angle. The efficiency of the proposed scheme has been evaluated by multiple simulation studies.

References

[1] S. Scheding, G. Dissanayake, E. M. Nebot, and H. Durrant-Whyte, "An experiment in autonomous navigation of an underground mining vehicle," *Robotics and Automation, IEEE Transactions on*, vol. 15, no. 1, pp. 85–95, 1999.

[2] J. Roberts, E. Duff, P. Corke, P. Sikka, G. Winstanley, and J. Cunningham, "Autonomous control of underground mining vehicles using reactive navigation," in *Robotics and Automation, 2000. Proceedings. ICRA'00. IEEE International Conference on*, vol. 4. IEEE, 2000, pp. 3790–3795.

[3] N. Nilsson, "A mobile automaton: An application of artificial intelligence techniques," DTIC Document, Tech. Rep., 1969.

[4] B. Chazelle, "Approximation and decomposition of shapes," *Advances in Robotics*, vol. 1, pp. 145–185, 1987.

[5] E. Guechi, J. Lauber, and M. Dambrine, "On-line moving-obstacle avoidance using piecewise bezier curves with unknown obstacle trajectory," in *Control and Automation, 2008 16th Mediterranean Conference on*. IEEE, 2008, pp. 505–510.

[6] K. Usher, "Obstacle avoidance for a non-holonomic vehicle using occupancy grids," in *2006 Australasian Conference on Robotics and Automation*, 2006.

[7] S. Ge and Y. Cui, "New potential functions for mobile robot path planning," *Robotics and Automation, IEEE Transactions on*, vol. 16, no. 5, pp. 615–620, 2000.

[8] I. Škrjanc and G. Klančar, "Optimal cooperative collision avoidance between multiple robots based on bernstein–bézier curves," *Robotics and Autonomous systems*, vol. 58, no. 1, pp. 1–9, 2010.

[9] J. Ng and T. Bräunl, "Performance comparison of bug navigation algorithms," *Journal of Intelligent & Robotic Systems*, vol. 50, no. 1, pp. 73–84, 2007.

[10] V. Lumelsky and A. Stepanov, "Dynamic path planning for a mobile automaton with limited information on the environment," *Automatic Control, IEEE Transactions on*, vol. 31, no. 11, pp. 1058–1063, 1986.

[11] I. Kamon, E. Rimon, and E. Rivlin, "Tangentbug: A range-sensor-based navigation algorithm," *The International Journal of Robotics Research*, vol. 17, no. 9, pp. 934–953, 1998.

[12] I. Kamon and E. Rivlin, "Sensory-based motion planning with global proofs," *Robotics and Automation, IEEE Transactions on*, vol. 13, no. 6, pp. 814–822, 1997.

[13] V. Lumelsky and T. Skewis, "Incorporating range sensing in the robot navigation function," *Systems, Man and Cybernetics, IEEE Transactions on*, vol. 20, no. 5, pp. 1058–1069, 1990.

[14] N. Buniyamin, W. Wan Ngah, N. Sariff, and Z. Mohamad, "A simple local path planning algorithm for autonomous mobile robots," *International journal of systems applications, Engineering & development*, vol. 5, no. 2, pp. 151–159, 2011.

[15] T. Nayl, G. Nikolakopoulos, and T. Guastafsson, "Kinematic modeling and simulation studies of a lhd vehicle under slip angles," in *Computational Intelligence and Bioinformatics/755: Modelling, Identification, and Simulation*. ACTA Press, 2011.

[16] T. Chaudhry, T. Gulrez, A. Zia, and S. Zaheer, "Bézier curve based dynamic obstacle avoidance and trajectory learning for autonomous mobile robots," in *Intelligent Systems Design and Applications (ISDA), 2010 10th International Conference on*. IEEE, 2010, pp. 1059–1065.